STIMO: An Interactive Robot for Learning and Gamification in BSCPE Education

A Thesis
Presented to the Faculty of
Information and Communications Technology Program
STI College Kalibo

In Partial Fulfilment
of the Requirements for the Degree
Bachelor of Science in in Computer Engineering

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November 22, 2025

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ABSTRACT

Title of research**:** *STIMO: An Interactive Robot for Learning and Gamification in BSCpE Education*

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Degree: **Bachelor of Science in Computer Engineering**

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STIMO, or An Interactive Robot for Learning and Gamification in BSCpE Education, is a learning assistant robot designed to make studying engaging, interactive, and enjoyable for students. Powered by a Raspberry Pi 4 single-board computer, STIMO integrates hardware and software to deliver three primary modes: STIMO Mode, Game Mode, and PC Mode. Additionally, all mini-games in STIMO are specifically tailored for BSCpE 3rd-year students, ensuring that every activity aligns directly with the lessons, handouts, and outcomes of the BSCpE 3-1 curriculum.

In STIMO Mode, the robot provides text-to-speech responses, animations, and a built-in calendar system for managing academic schedules. Game Mode transforms lessons into interactive mini-games aligned with the BSCpE 3-1 curriculum, including Hangman, Word Scramble, Fill-in-the-Blank Challenge, and Odd One Out. PC Mode functions as a Raspberry Pi desktop environment, allowing access to applications and files while STIMO continues to respond to voice commands in the background.

The system aims to enhance student motivation, productivity, and engagement by combining robotics, gamification, and educational tools. Through an integration of servo motor movements, voice interaction, fingerprint authentication, and touch-based input, STIMO demonstrates a holistic approach to human-computer interaction in education. Although limited by processing power, internet dependency, and battery life, STIMO serves as a prototype that bridges technology and learning through play, personalization, and accessibility.

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INTRODUCTION

STIMO, or An Interactive Robot for Learning and Gamification in BSCpE Education, is a learning assistant robot designed to make studying engaging, interactive, and enjoyable for students. Powered by a Raspberry Pi 4 single-board computer, STIMO integrates both hardware and software into one cohesive system that provides three distinct modes tailored to user needs: STIMO Mode, Game Mode, and PC Mode.

STIMO Mode features preprogrammed responses synchronized with servo motor movements and animated facial expressions, creating an engaging and lifelike interaction experience. It includes a built-in calendar system where students can input homework schedules through touch or keyboard, stored in a local SQLite database, with automated reminders delivered through TTS, expressive animations, and servo gestures. Students can casually talk to STIMO for simple questions or light emotional support, and for informational queries, STIMO uses Wikipedia and Google Search APIs, with custom filtering code that turns the retrieved information into short, natural-sounding responses spoken in STIMO's tone. Additional interactive features such as face detection allow STIMO to greet the user or react when someone appears in front of the camera, and even while students work in PC Mode or Game Mode, STIMO Mode continues running in the background, enabling voice-activated interaction—complete with synchronized speech, animations, and servo movements—without interrupting the user's workflow.

Game Mode transforms academic learning into an enjoyable and interactive experience by integrating subject-specific mini-games. Tailored for the BSCpE 3-1 curriculum, this mode offers a structured learning flow: students can choose from eight subjects, select a grading period (Prelim, Midterms, Prefinals, or Finals), and play a mini-game corresponding to that period. For instance, Hangman is featured in the Prelim period, Word Scramble for Midterms, Fill-in-the-Blank Challenge for Prefinals, and Odd One Out for Finals. Each game is linked to specific handouts for the chosen subject and period, promoting an engaging and curriculum-based review experience.

PC Mode allows STIMO to function as a fully operational Raspberry Pi desktop system,

enabling tasks such as accessing files, using applications, and completing assignments. While in PC Mode, STIMO Mode continues to run in the background, ready to receive input through GPIO-connected buttons. Users can activate the microphone for speech-to-text or display an on-screen keyboard for manual typing. This setup provides an interactive and accessible experience within the Raspberry Pi environment, allowing users to control STIMO's functions through button inputs and optional voice input.

Background of the problem

Recent studies highlight the growing challenges faced by students in maintaining motivation and engagement with academic tasks. According to the Healthy Minds Study, 44% of college students across 133 campuses reported experiencing symptoms of depression in 2021–2022, with 15% considering suicide due to overwhelming academic and social pressures. These issues often stem from the lack of interactive learning tools, with students finding traditional methods monotonous and unengaging, which negatively impacts their academic performance and mental well-being. [KBW, 2023]

Furthermore, research from the University of Michigan School of Public Health indicates that 67% of students report improved engagement and motivation when learning incorporates interactive and gamified elements. However, many existing platforms—such as learning management systems—do not provide the level of hands-on interactivity needed to keep students consistently engaged. This gap underscores the need for innovative tools like STIMO, which aims to enhance learning through game-based activities, personalized feedback, and interactive features. [JTH, 2023]

Overview of the current state of the technology

The current state of educational technologies, such as the eLMS at STI College, effectively support students with structured learning, offering essential features like grade tracking, content delivery, and communication tools. However, there is a growing need for more engaging, interactive platforms that can further enhance student motivation and provide immersive learning experiences. This is where STIMO comes in. By integrating both hardware and software, STIMO offers a dynamic, interactive study companion that not only enhances learning engagement but also helps students stay motivated and organized with personalized reminders and gamified study tools.

STIMO's Unique Features in Gamification and Robotics

STIMO offers innovative features that set it apart from existing educational tools and robots like Kahoot! Quizlet, Nao, and Pepper. Unlike generic platforms, STIMO focuses on the BSCpE 3-1 curriculum, ensuring relevance and engagement for students.

Features of STIMO

Subject-Specific Mini-Games:

STIMO includes games tailored to the BSCpE curriculum, such as Hangman (Prelim), Word Scramble (Midterms), Fill-in-the-Blank Challenge (Prefinals), and Odd One Out (Finals). Each game is linked to specific subjects and handouts, making studying interactive and relevant.

Course-Specific Integration:

STIMO's mini-games and features are designed to align with actual modules and topics from the BSCpE 3-1 program, providing more accurate and useful learning compared to general quiz apps.

Motorized Movements and Animations:

STIMO enhances learning by incorporating physical movements and facial animations synchronized with its interactive features, making it a personalized study companion.

Personalized Educational Support:

STIMO offers students a tailored learning experience, combining gamified study tools with course-specific content to improve engagement and understanding.

Holistic Approach:

By merging hardware components like servo motors and face animations with educational mini-games, STIMO provides a fun, engaging, and effective study tool designed specifically for STI College students

Objectives of the study

General Objective

To design and develop STIMO, an interactive and engaging robot that combines hardware and software, made learning more enjoyable for STI College students

Specific Objectives

To develop a Robot That Synchronizes Motorized Limb Movements and Preprogrammed Responses with Animations.

STIMO integrates motorized limb movements, animated facial expressions, and preprogrammed responses using MG90S servo motors controlled by the Raspberry Pi Pico through a PCA9685 PWM driver module. These movements are synchronized in real time with the Raspberry Pi 4's processing of text-to-speech outputs and on-screen facial animations, ensuring fluid and coordinated interaction between STIMO's limb movements, visual expressions, and verbal responses.

To create a Game Mode That Incorporates Subject-Specific Mini-Games Tailored to the BSCpE 3-1 Curriculum and Handout

Game Mode offers mini-games tailored to the BSCpE 3-1 curriculum, including Hangman, Word Scramble, Fill-in-the-Blank Challenge, and Odd One Out, aligned with specific subjects and periods (Prelim, Midterms, Prefinals, Finals). This gamified approach transforms traditional studying into an interactive and engaging experience

To develop a Calendar System That Allows Students to Input, View, and Manage Their Schedules.

The built-in calendar allows students to add, view, and manage schedules through the dedicated icon that you can touch with the lcd's touchscreen on STIMO's face side. or speech-to-text input or keyboard input. Events are stored in a database, and STIMO provides reminders using animations, movements, and text-to-speech notifications

To Implement Push Buttons for Seamless Switching Between STIMO's Modes

MCU-based push buttons allow users to switch smoothly between STIMO Mode, Game Mode, and PC Mode. These buttons are connected to the Raspberry Pi GPIO pins, When a mode button is pressed, the system closes the current interface and launches the corresponding mode's Python program. Even while PC Mode is active, STIMO Mode continues running in the background as an overlay, ready to respond through speech-to-text or the on-screen keyboard.

Scope and limitations of the study

It Uses Fingerprint Authentication for Security

The fingerprint sensor operates through a separate standalone security program that launches automatically when the Raspberry Pi boots. This authentication system is independent from STIMO and does not rely on Windows or any built-in OS lock features. The program verifies the user's identity using its own fingerprint database and also provides fallback options such as password entry. Only after successful authentication does the system allow access to STIMO and other modes.

It Utilizes a Microphone and Speakers for Input and Output

The microphone captures user commands using speech-to-text (STT) functionality, while the speakers provide clear text-to-speech (TTS) responses. These components work together to ensure effective two-way communication in STIMO Mode.

It Has MCU Buttons for Input Selection

The GPIO-connected push buttons allow users to switch between input methods, such as speech-to-text for voice commands or the on-screen keyboard for typing. These buttons are directly managed by the Raspberry Pi 4 for button handling ensuring smooth, responsive, and real-time interaction.

It Includes a Portable Power Bank and Battery

The power bank and battery provides energy for all components, ensuring STIMO's portability and extended operational time. It integrates seamlessly with the DCDC-

NUC converter to supply consistent power.

STIMO MODE

Text-to-Speech (TTS) Responses:

Provides verbal responses to user commands using preprogrammed phrases and text-to-speech technology

Animated Facial Expressions:

Displays dynamic and responsive facial animations including three emotion-based background color changes: yellow for happy, blue for sad, and red for angry. These facial emotions are triggered by STIMO's preprogrammed responses and casual talking commands, adding a layer of expressiveness.

Servo Motor Movements:

Performs synchronized physical movements, such as kicking or waving, using MG90S Servo Motors controlled by the Raspberry Pi Pico.

Interactive Calendar System:

Allows students to add, view, edit, and delete homework schedules. Students can check homework due today, view homework for the next day, or see all upcoming assignments. The system provides spoken announcements and visual animations when displaying homework information, helping students stay organized and on

track with their academic responsibilities.

Always-Active Background Functionality:

Operates in the background to listen for commands while other modes are active, ensuring seamless multitasking.

It Has a Camera Module for Visual Interaction:

The Raspberry Pi Camera Module allows STIMO to detect faces and user presence in real time, enabling greetings, animations, and simple vision-based interactions.

GAME Mode

Game Mode integrates educational mini games tailored to the BSCpE 3- 1 curriculum. Students can choose from eight subjects, with each subject divided into four terms: Prelim, Midterms, Prefinals, and

Finals. Each term offers a specific game linked to subject handouts:

- Prelim: Hangman (Word Guessing Game)
- Midterms: Word Scramble
- Prefinals: Fill-in-the-Blank Challenge
- Finals: Odd One Out

These games provide an interactive learning experience, helping students review lessons while maintaining engagement and focus

Gaming Mode Switching Behavior:

When the user leaves Gaming Mode, the current game session is intentionally terminated to prevent background resource usage and potential conflicts with other modes. Gaming Mode remains inactive until the user selects it again. Upon returning, the system relaunches the game interface from the beginning, ensuring stability and consistent performance.

PC MODE

PC Mode transforms STIMO into a fully functional Raspberry Pi OS desktop, enabling traditional computing tasks such as file management, web browsing, and document creation. It ensures multitasking by keeping STIMO Mode active in the background, allowing the user to interact with it through voice commands or on-screen keyboard input. This mode balances productivity, accessibility, and interactivity within a Linux-based environment.

Raspberry Pi OS Environment: *Provides a complete desktop interface for productivity tasks such as browsing, managing files, and creating documents.*

Multitasking Capability: *Allows users to switch seamlessly between PC Mode, Game Mode, and STIMO Mode using GPIO-connected buttons, with STIMO Mode remaining active in the background.*

Speech-to-Text (STT) and Keyboard Input Support: *Enables users to communicate with STIMO through voice commands or on-screen keyboard input while working in PC Mode.*

Background Assistant Functionality: *Operates STIMO Mode as a minimized overlay, ready to perform tasks such as conducting searches or launching applications while users utilize the Raspberry Pi desktop*

Limitations of the Study

It requires an internet connection

The Raspberry Pi 4 Model B powering STIMO is optimized for lightweight multitasking and educational applications but is not designed to handle highly demanding software. Resource-intensive programs such as advanced 3D modeling tools (e.g., Blender or AutoCAD) or professional video editing suites (e.g., Adobe Premiere Pro) require significantly greater processing power, memory, and graphics performance than what the Raspberry Pi 4 can provide. As a result, STIMO is best suited for interactive learning, gamification, and productivity tasks rather than computationally intensive workloads.

It lacks high-performance processing

The Raspberry Pi 4 Model B powering STIMO is optimized for lightweight multitasking and educational applications but is not designed to handle highly demanding software. Resource-intensive programs such as advanced 3D modeling tools (e.g., Blender or AutoCAD) or professional video editing suites (e.g., Adobe Premiere Pro) require significantly greater processing power, memory, and graphics performance than what the Raspberry Pi 4 can provide. As a result, STIMO is best suited for interactive learning, gamification, and productivity tasks rather than computationally intensive workloads.

It has limited battery runtime

STIMO's 6000mAh power bank provides approximately 1 hours of continuous operation, even under typical use. This limited runtime is often insufficient for extended sessions or high-power activities, necessitating frequent recharging to maintain functionality.

It is strictly limited to 3rd-year college students

The research was strictly limited to 3rd-year college students. This means that all findings, conclusions, and implications drawn are exclusive to the experience and feedback of this third-year population.

Review of related literature/studies/systems

Rasp-board: Digital Bulletin Board Using Raspberry Pi in Wireless Technology

Rasp-board uses Raspberry Pi and wireless technology to modernize communication through easy-to-use, efficient bulletin boards. While Rasp-board focuses on announcement systems, STIMO adapts similar technologies to create a more personalized assistant for

students, combining calendar reminders and animations for enhanced interaction. [ARA, 2023] Futronic FS80H USB 2.0 Fingerprint Scanner

This biometric device demonstrates robust and reliable integration of hardware with software, suitable for identity verification. While STIMO is not a biometric system, its integration with external tools like SQLite for calendar tasks and Windows for system-level actions reflects the seamless adaptability of hardware for diverse applications. [CLD, 2023]

Automated Waste Segregator: A Prototype for Improved Waste Management

This project uses Arduino Uno and buttons for efficient task management, such as controlling bins for sorting waste. STIMO adopts a similar MCU-based approach for task control, enabling users to switch modes like Game Mode and STIMO Mode through hardware inputs, ensuring simplicity and reliability. [JMR, 2022]

Design, Development, and Implementation of Educational and Entertainment Mobile Robots Utilizing Arduino Microcontroller

This study highlights how Arduino microcontrollers were used to blend education and entertainment through buttons for interaction. Similarly, STIMO combines learning and fun with curriculum-based games, providing a unique user experience through hardware-driven interactivity. [CRJ, 2018]

Acceptability and Relevance of an Innovated Arduino-Based Microcontroller Intrusion Alarm System.

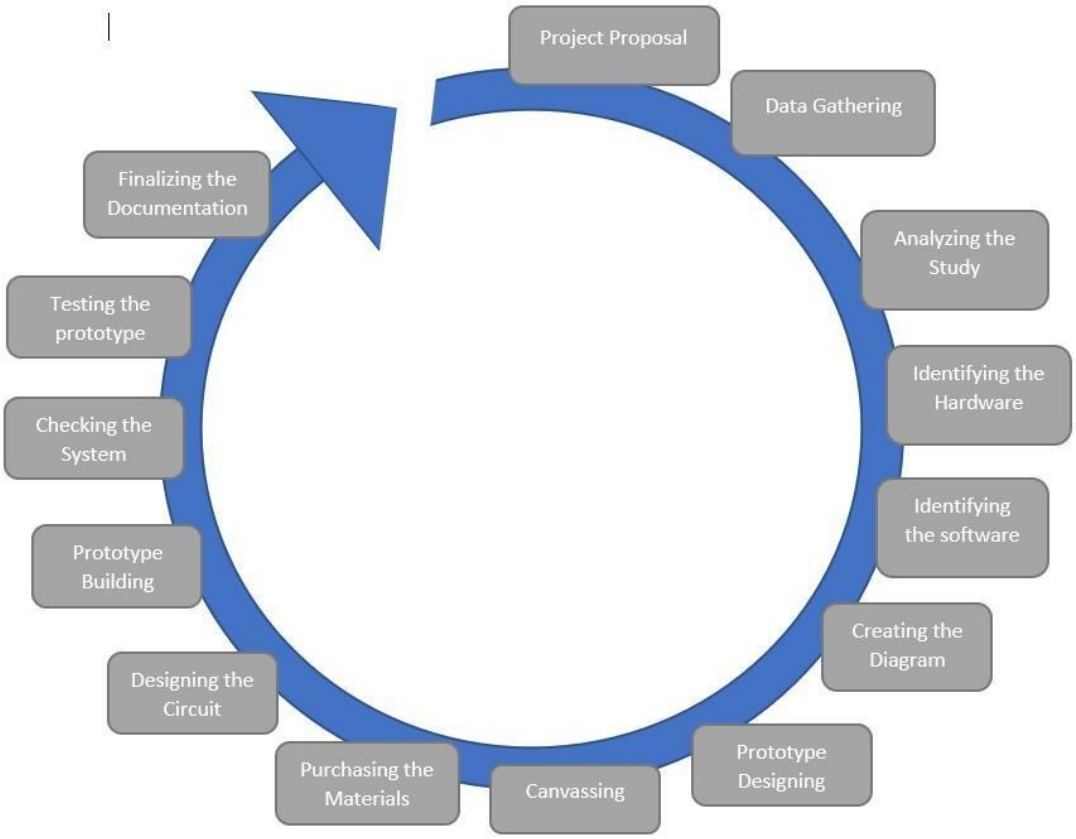
This project illustrates the use of Arduino microcontrollers and buttons for managing security functions. STIMO builds this concept by implementing intuitive button-based interactions to switch modes and facilitate various tasks, combining user-friendly controls with interactive learning and gaming systems. [VOL, 2022]

Synthesis

The reviewed literature highlights the integration of Arduino-based systems in creating interactive devices, such as game controllers and robotic companions, to improve user engagement through customizable features. This is particularly relevant for STIMO's hardware-driven design, which leverages servo motors and sensors to produce animated movements and physical interactions. Studies on gamification emphasize how well-designed games can enhance learning, which aligns with STIMO's curriculum-based mini-games tailored for BSCpE students. Additionally, examples of task-specific applications in local research, such as calendar management and automated systems, reinforce STIMO's practical approach to organizing student schedules while promoting interactive and engaging study experiences. STIMO bridges the gap between academic support and immersive interaction, combining physical robotics with software-driven learning tools.

METHODOLOGY

This subchapter details the methodology applied in the development of our STIMO project. We utilized a prototyping approach, which guided us throughout the creation process. The development involved iterative testing and adjustments until the optimal design and functionality were achieved. The procedure encompasses a series of stages listed below, representing a cyclical process that includes hardware and software integration, system evaluation, and refinement to meet the project's requirements effectively.



System design specification

Stage 1: Project Proposal

Defining the project concept, goals, and requirements.

Stage 2: Data Gathering

Researching relevant hardware, software, and educational tools to inform the design.

Stage 3: Analyzing the Study

Refining objectives and finalizing technical specifications based on gathered data.

Stage 4: Hardware Identification

Selecting essential components, including MCU-based controllers, sensors, and interactive displays.

Stage 5: Software Development

Developing the system's software, including predefined commands, emotes, and sensor integrations.

Stage 6: Circuit and System Design

Creating diagrams and schematics for the integration of hardware components.

Stage 7: Prototype Construction

Assembling the physical prototype with selected components and 3D-printed casing

Stage 8: Testing and Refinement Evaluating the system's functionality and making iterative improvements to meet the study's goals.

Data Flow

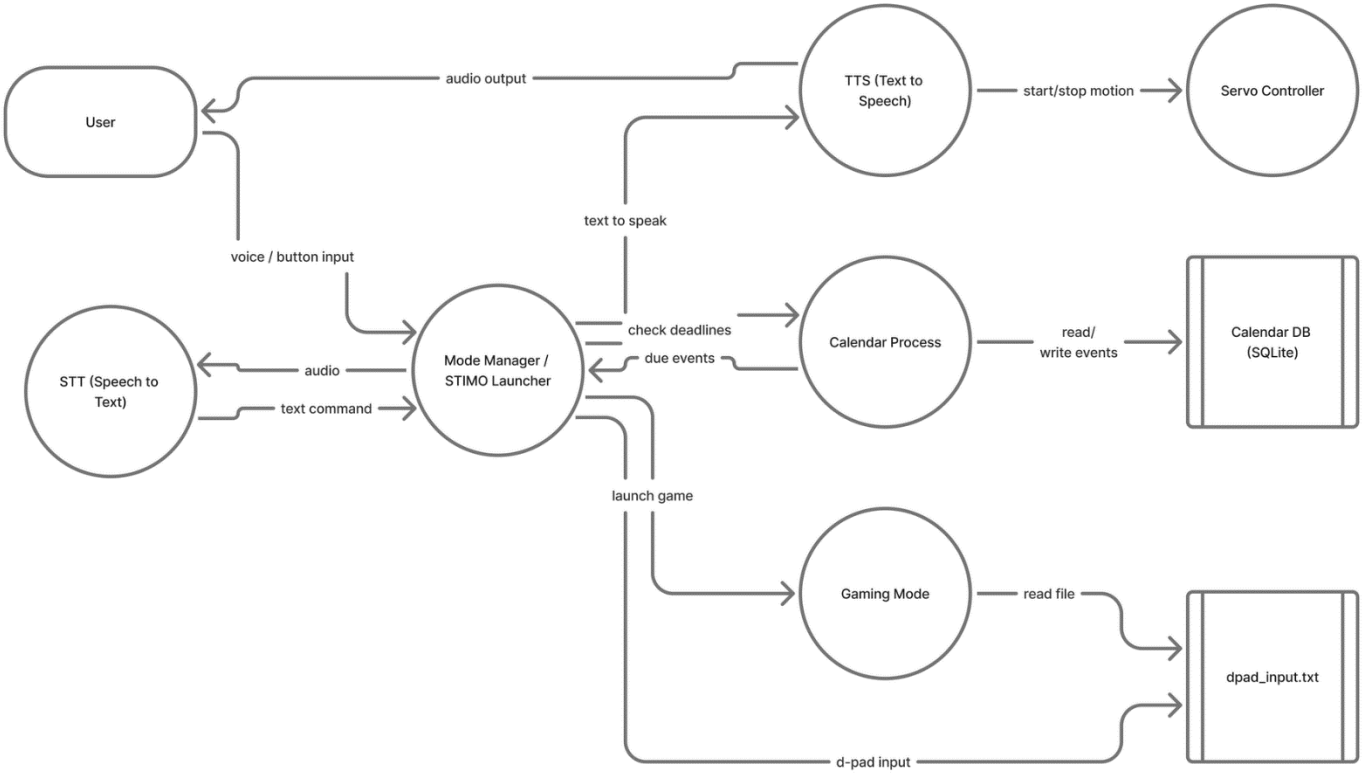


Figure 2:

Hardware Flowchart

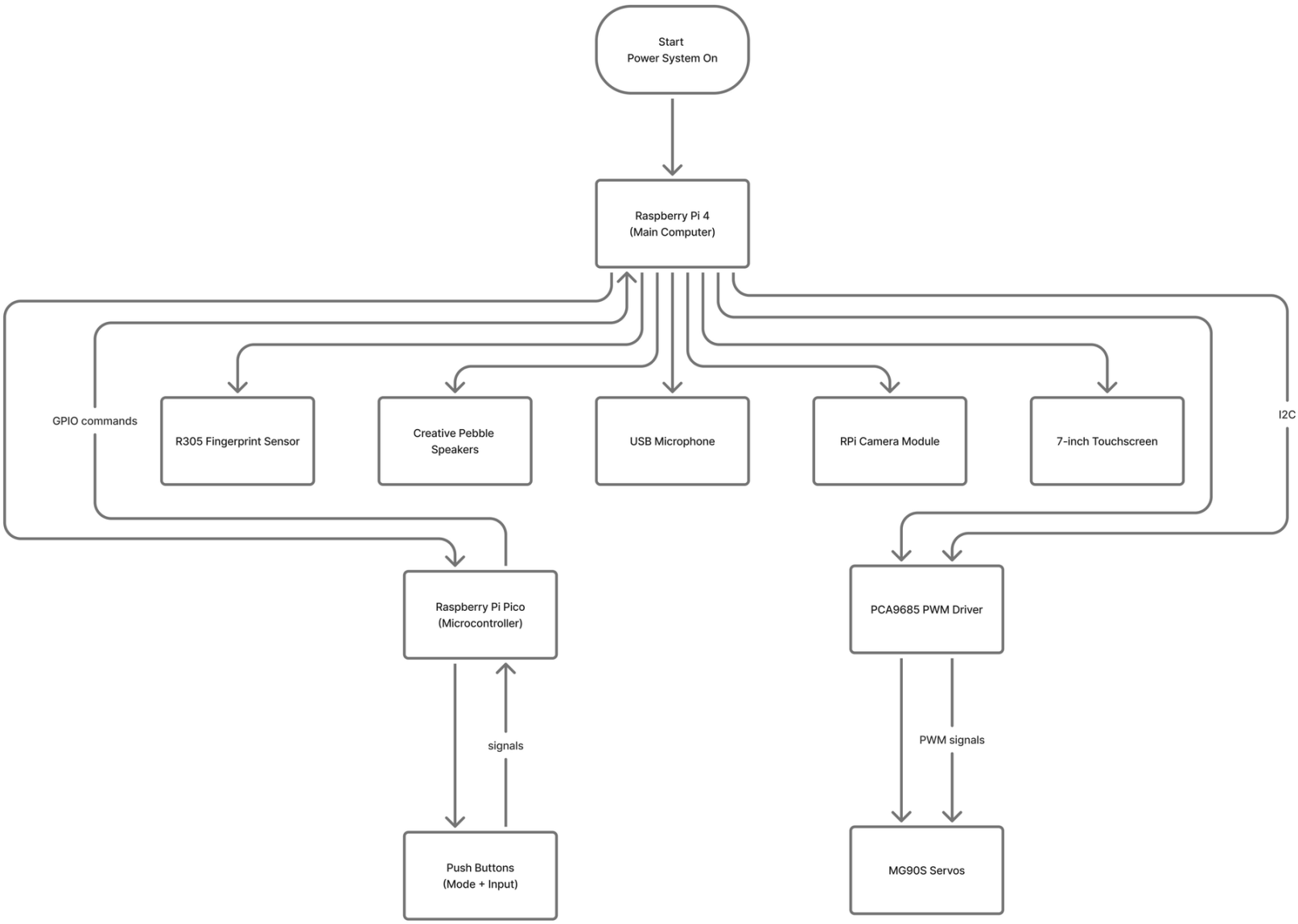


Figure 3:

Software Flowchart

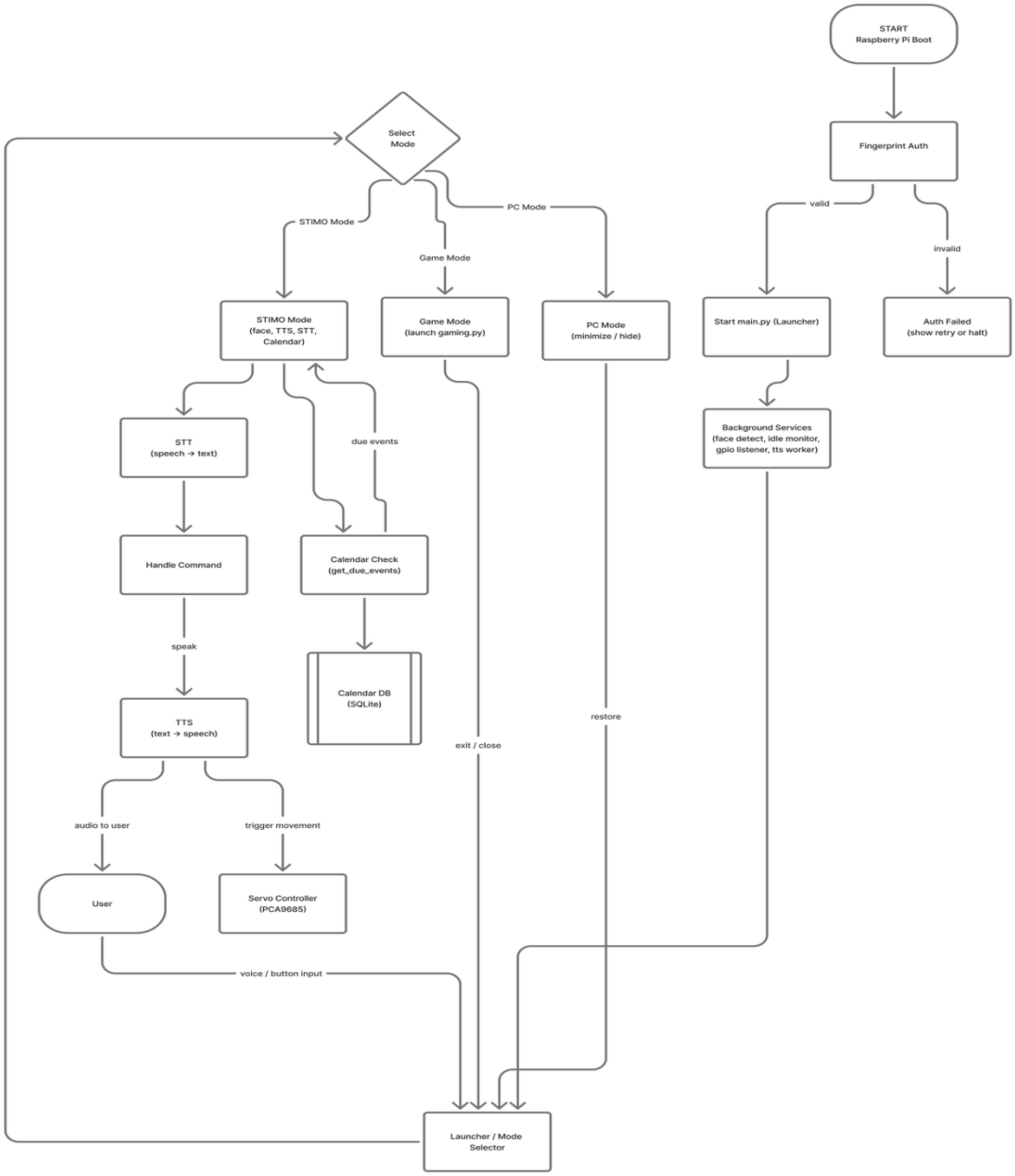


Figure 4:

Block Diagram

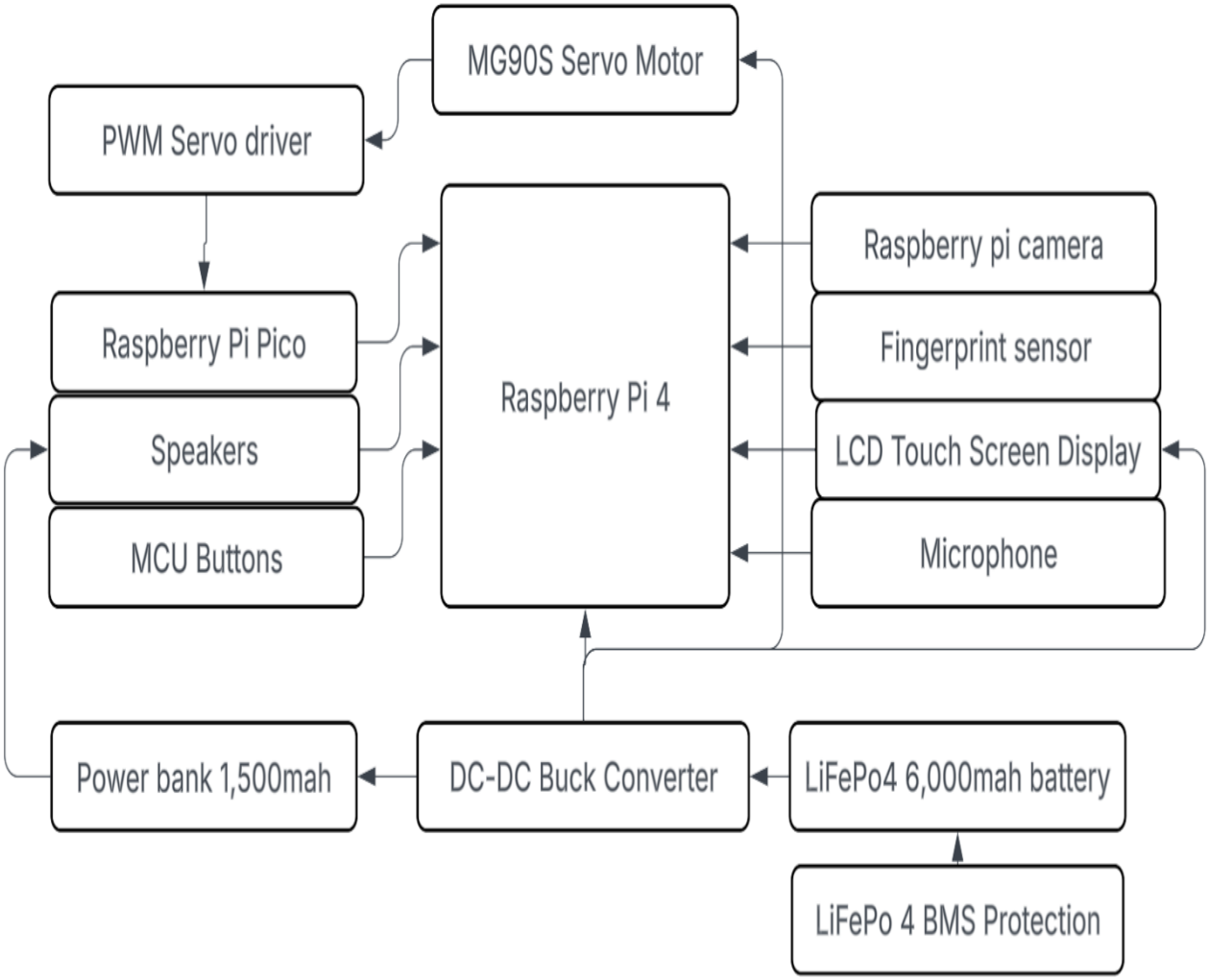


Figure 5:

Prototype Model

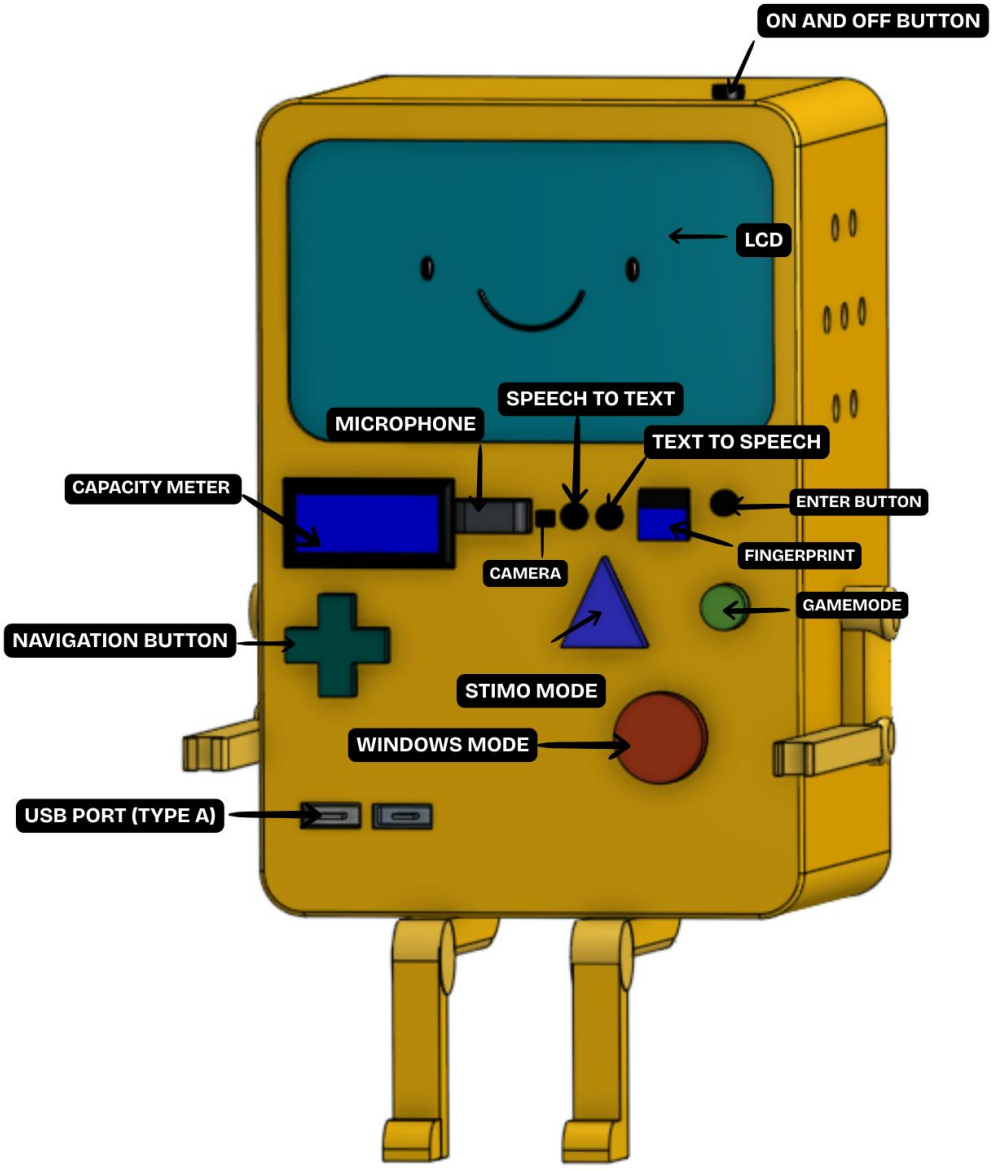


Figure 6: Model Design

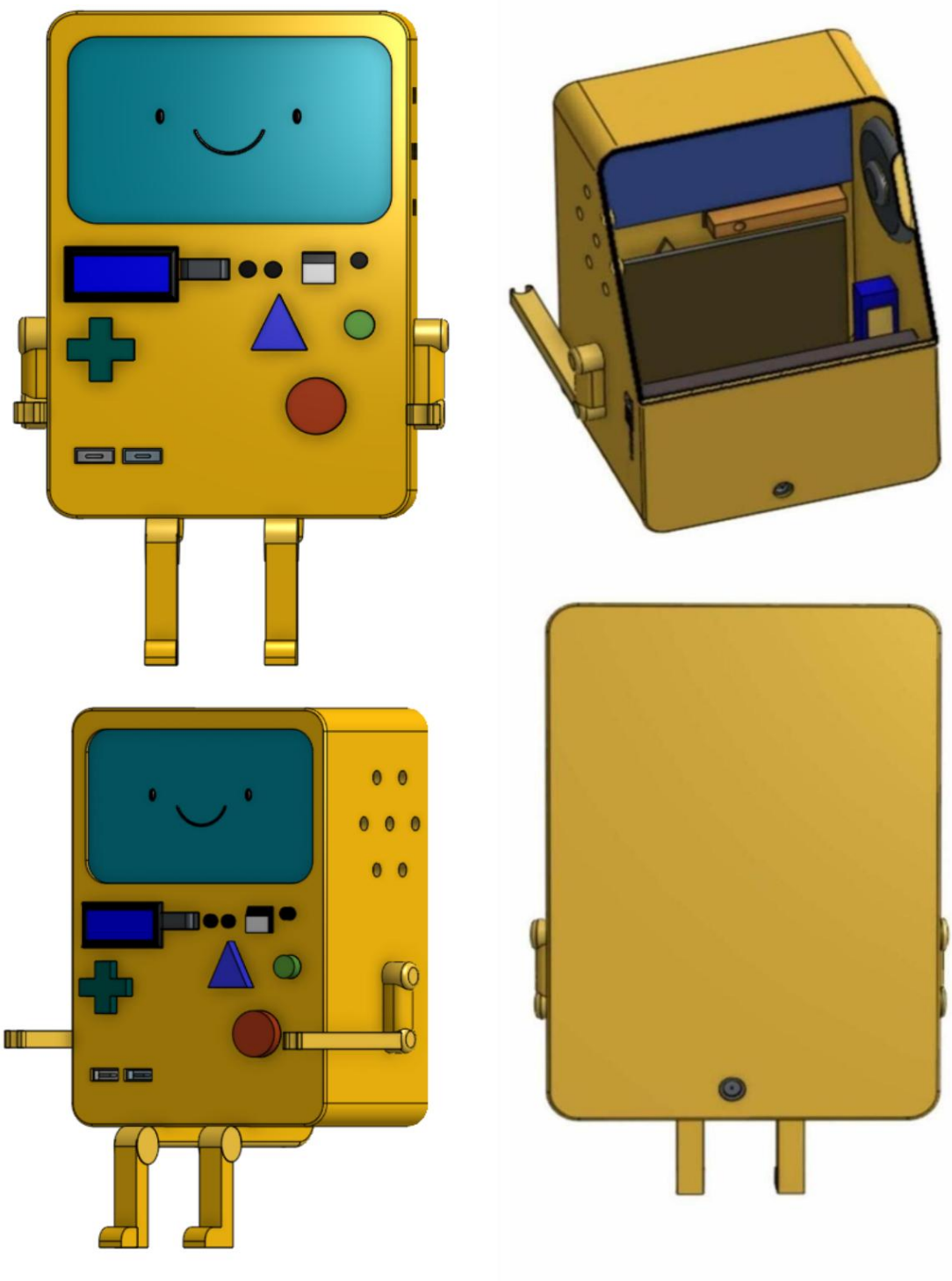


Figure 6.1 : Model Design

Fingerprint Authentication

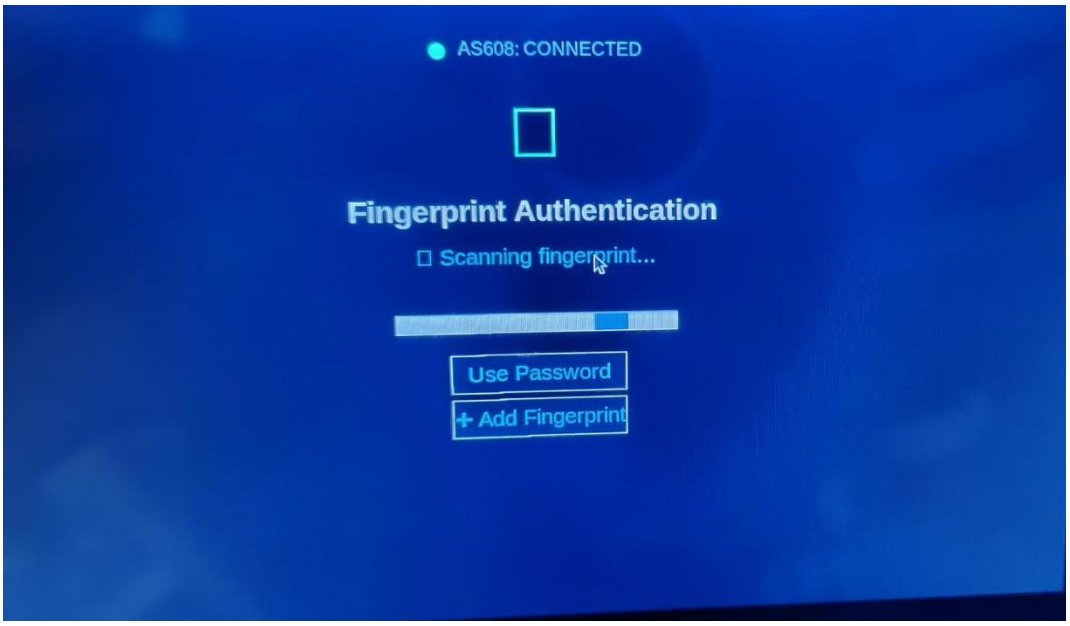


Figure 7: Fingerprint Authentication Scanning

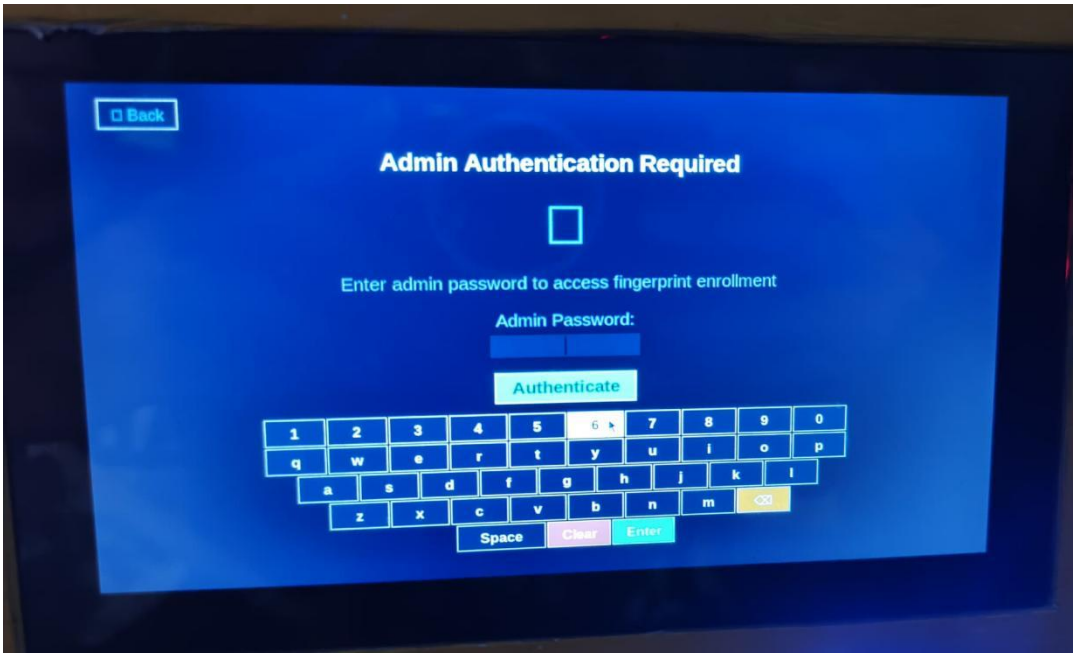


Figure 7.1: Admin Authentication Required

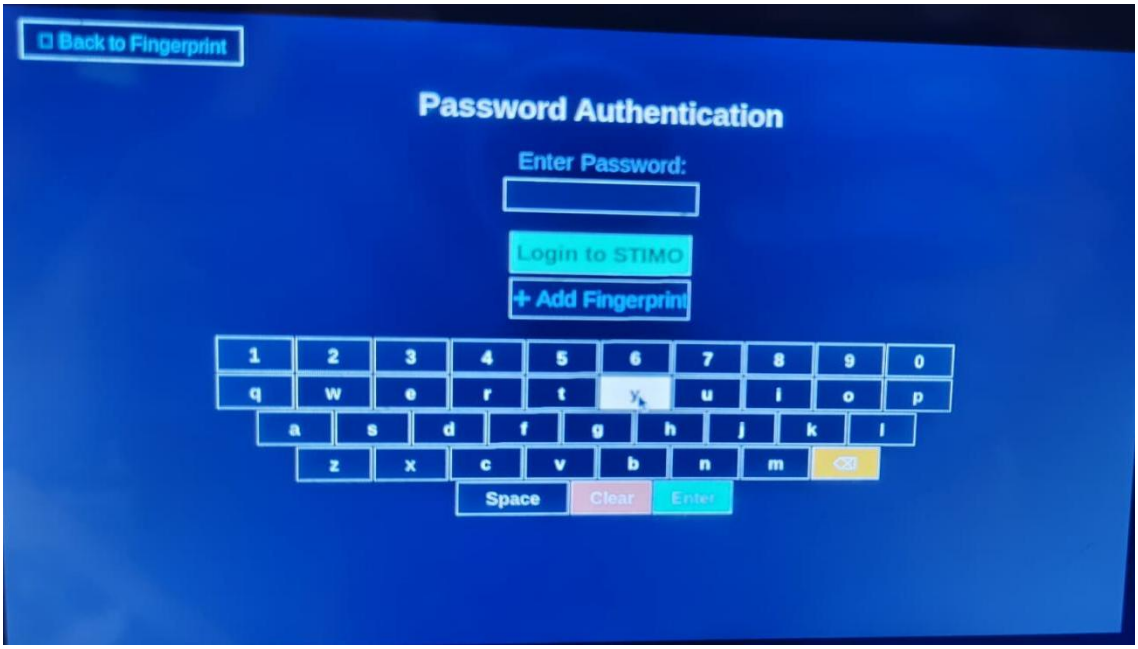


Figure 7.2: Password Authentication

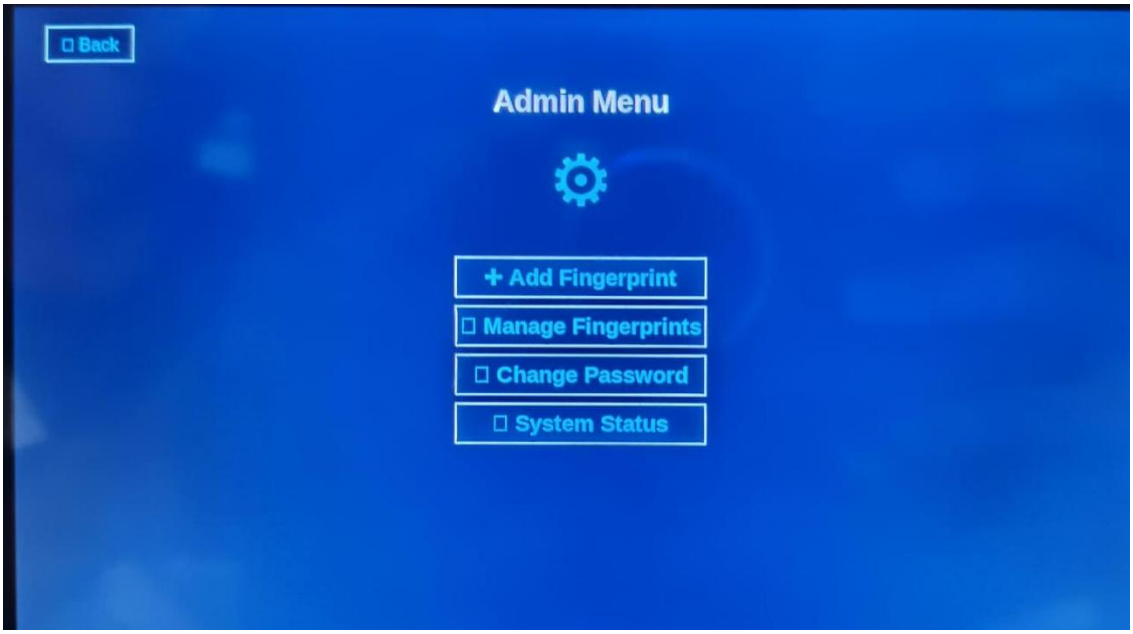


Figure 7.3: Admin Menu

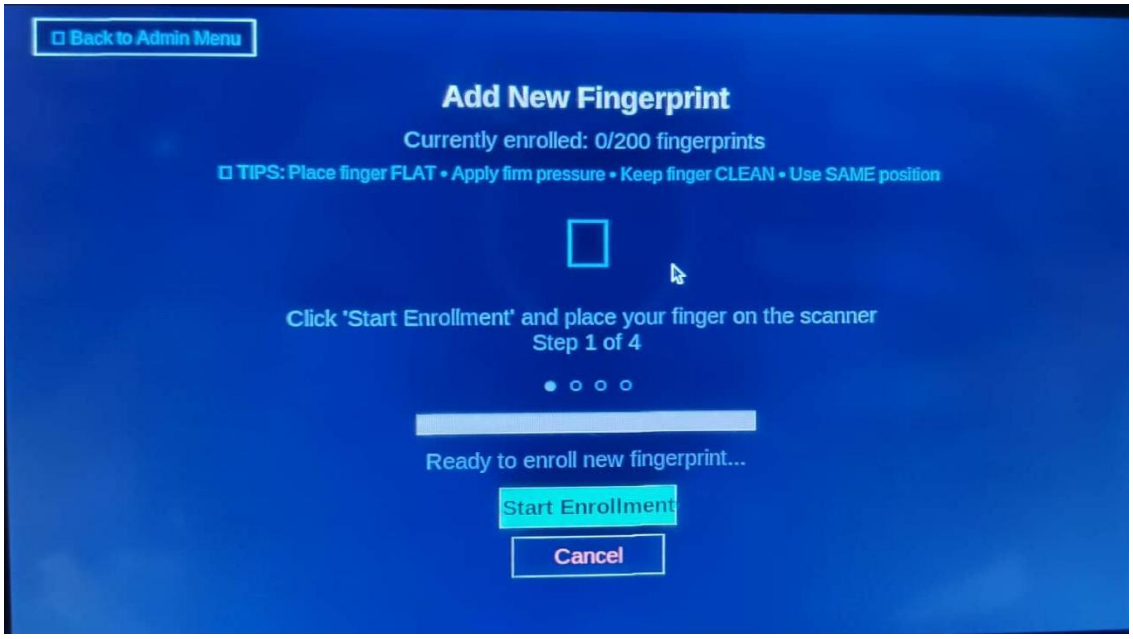


Figure 7.4: Add New Fingerprint

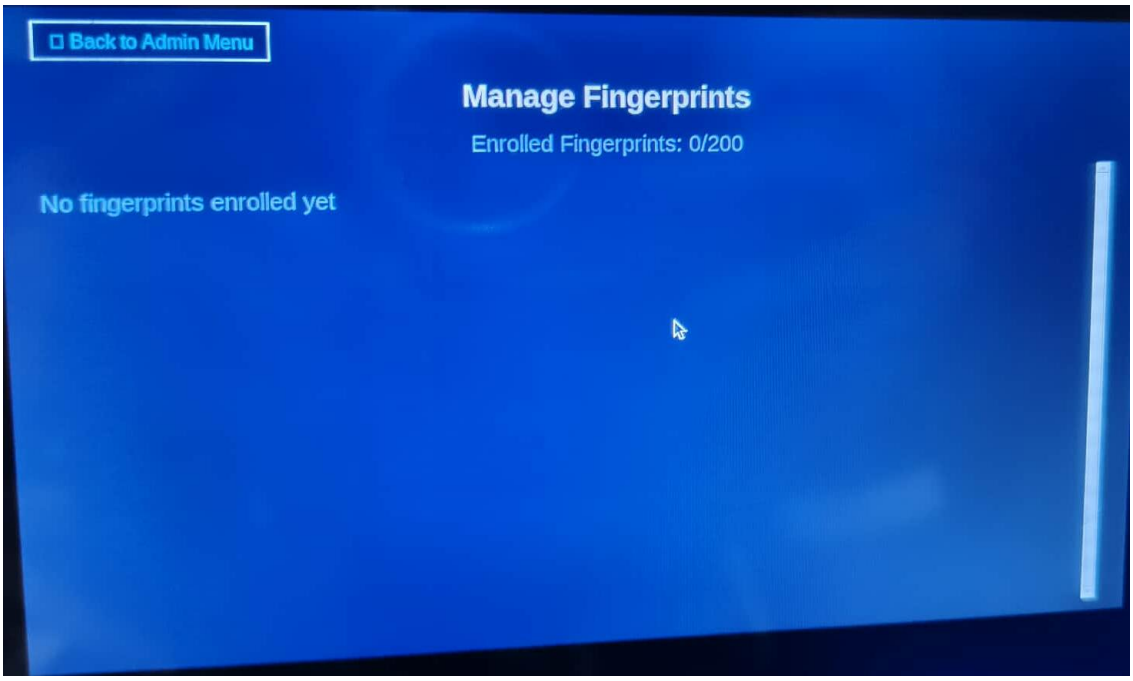


Figure 7.5: Manage Fingerprint

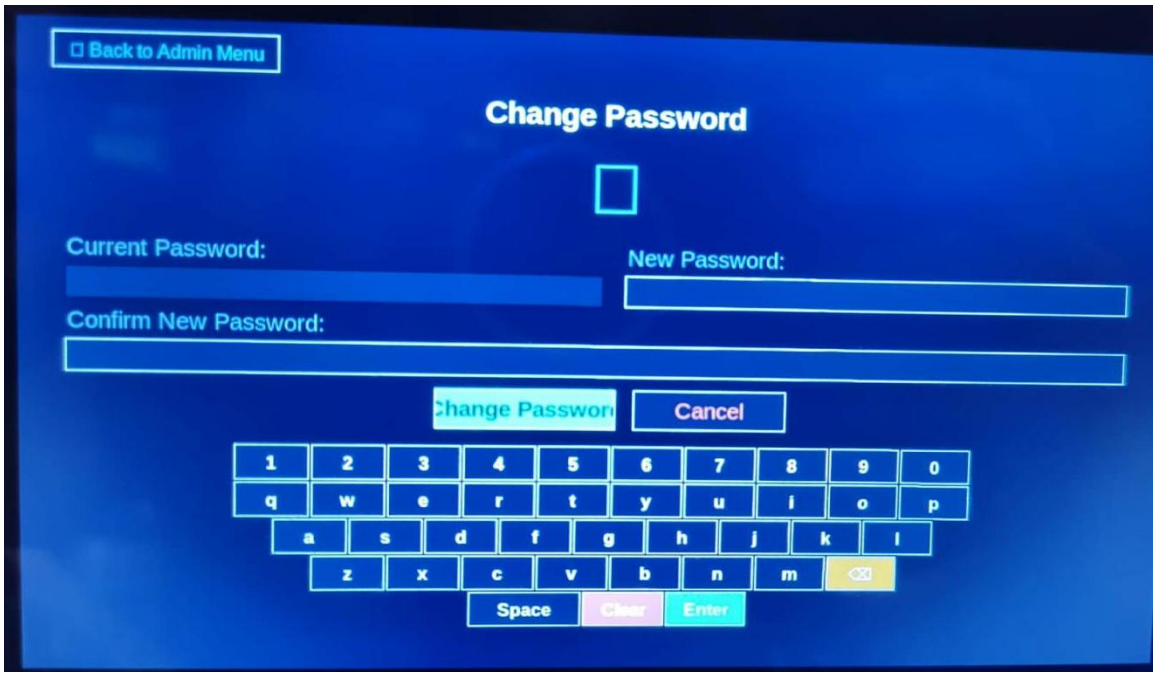


Figure 7.6: Change Password

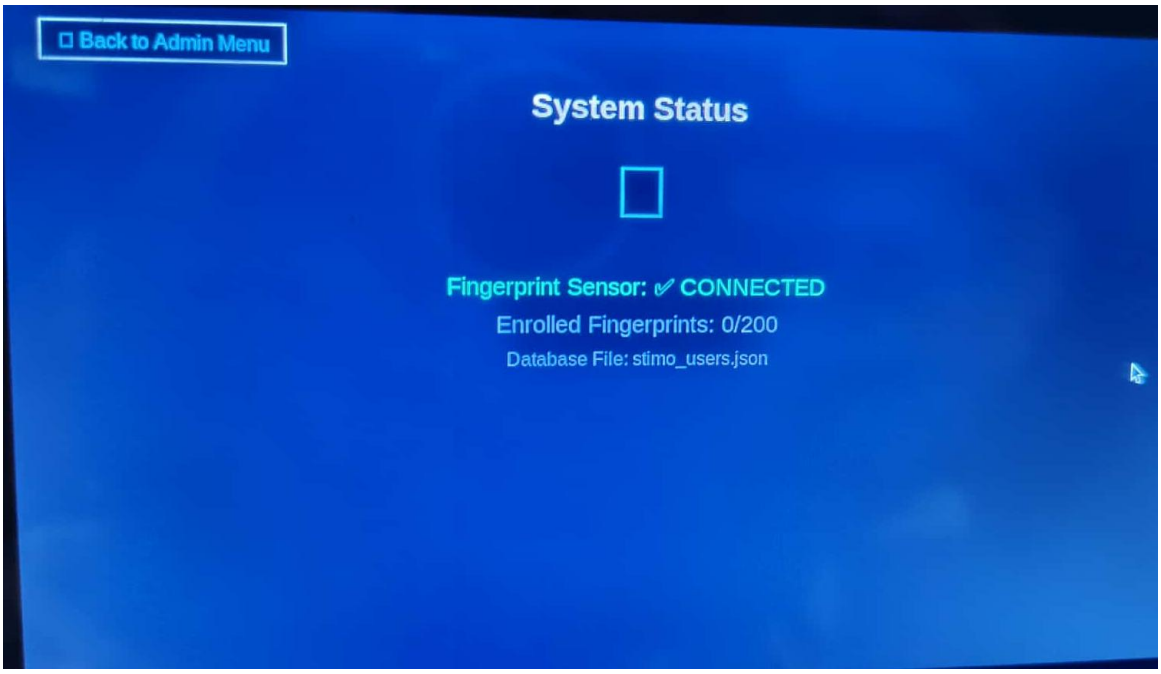


Figure 7.8: System Status

Launcher

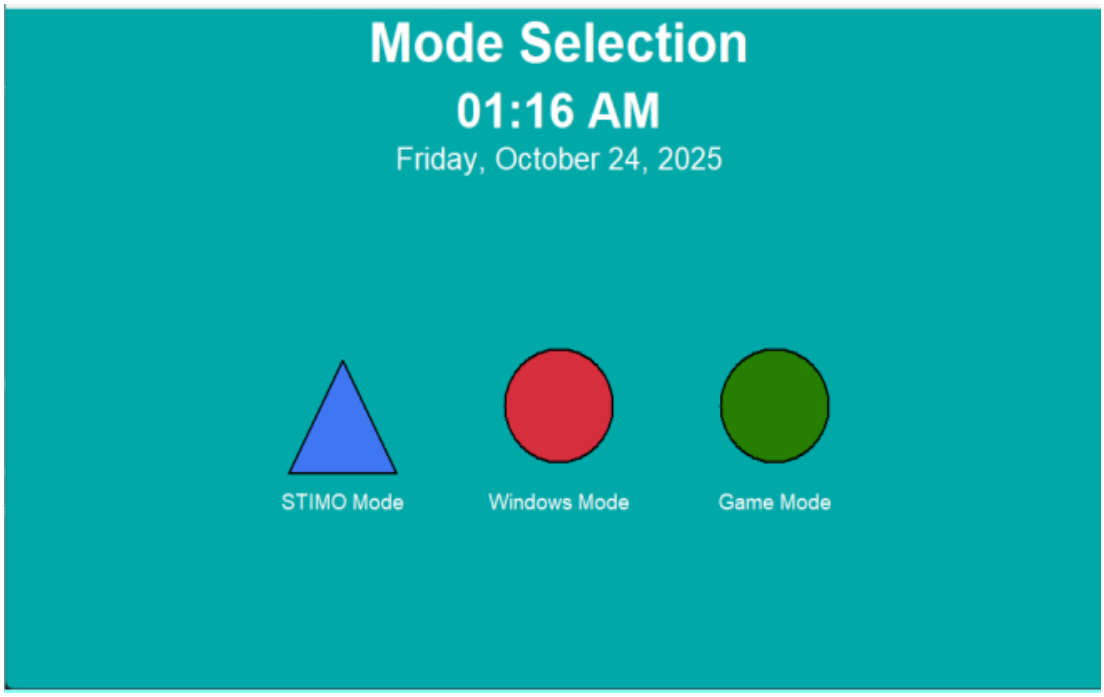


Figure 8: Mode Selection

Stimo Mode

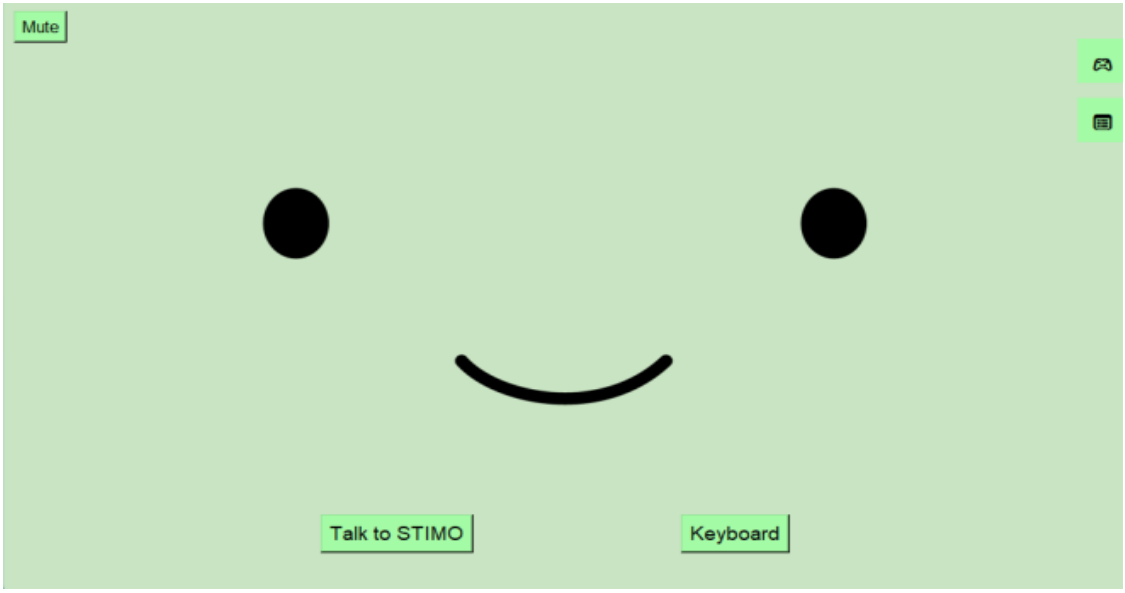


Figure 9: Stimo Mode Select

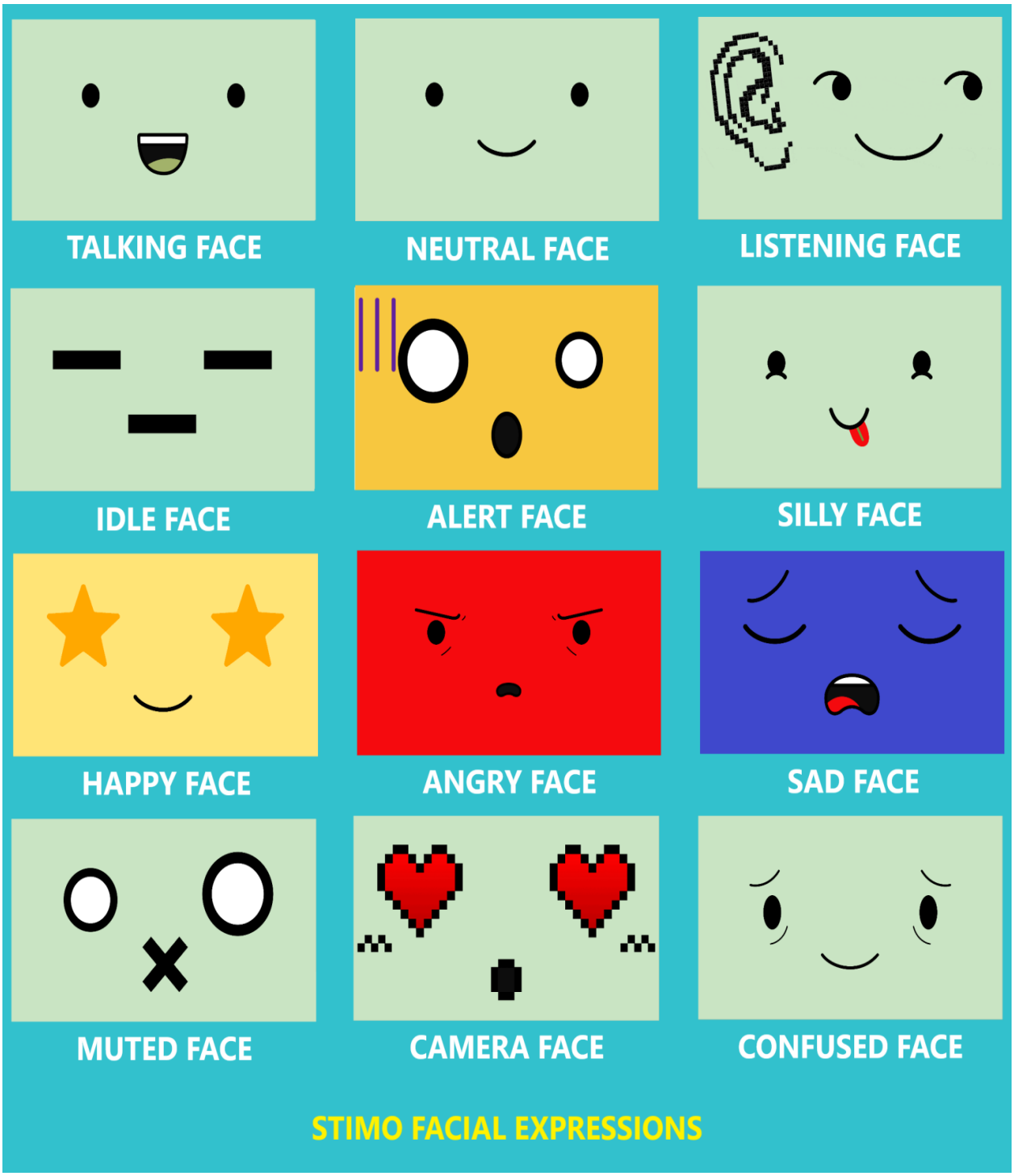


Figure 9.1: Stimo Facial Expressions

Stimo Calendar

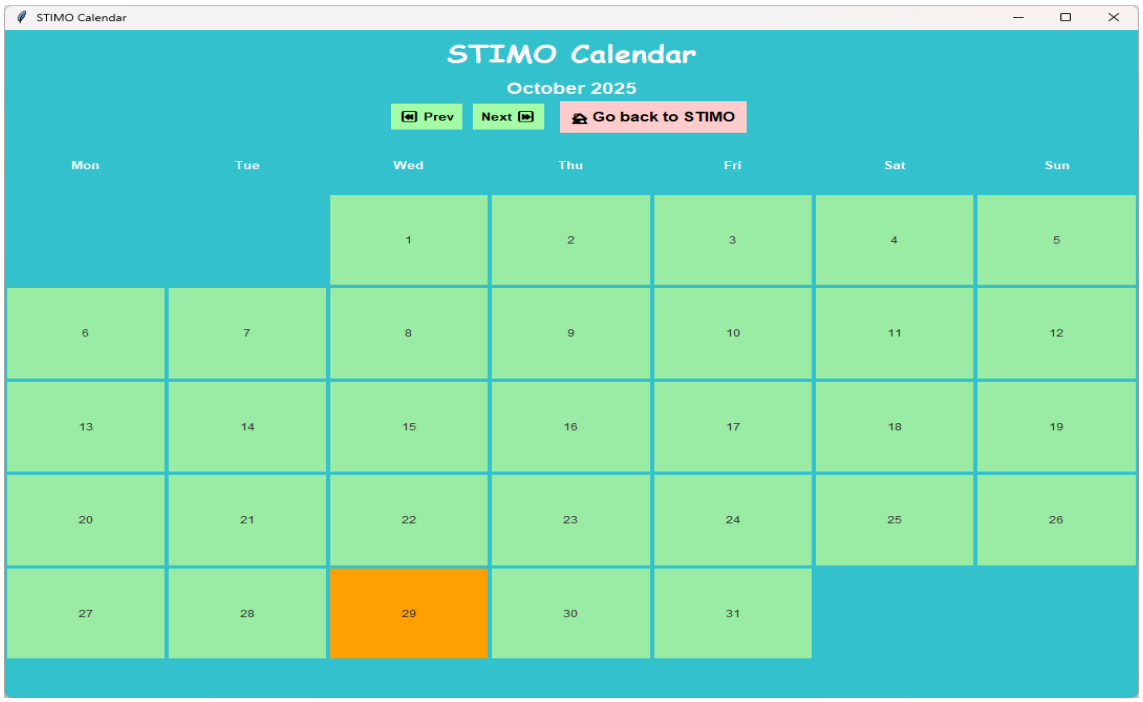


Figure 9.2: Stimo calendar

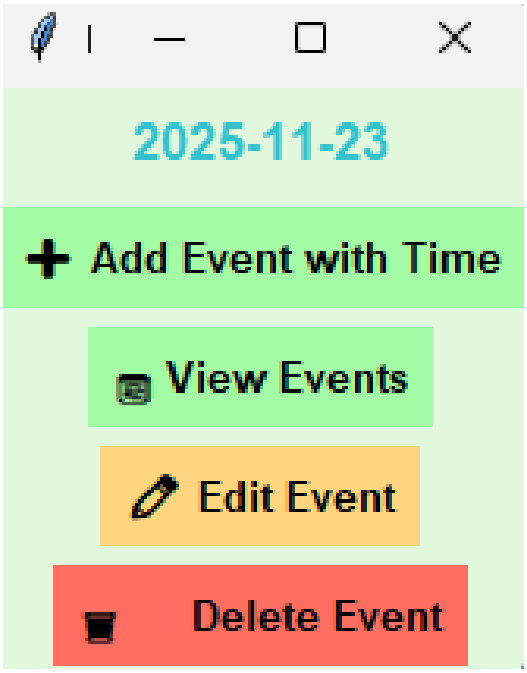


Figure 9.3: Selection Add,View,Edit,Delete

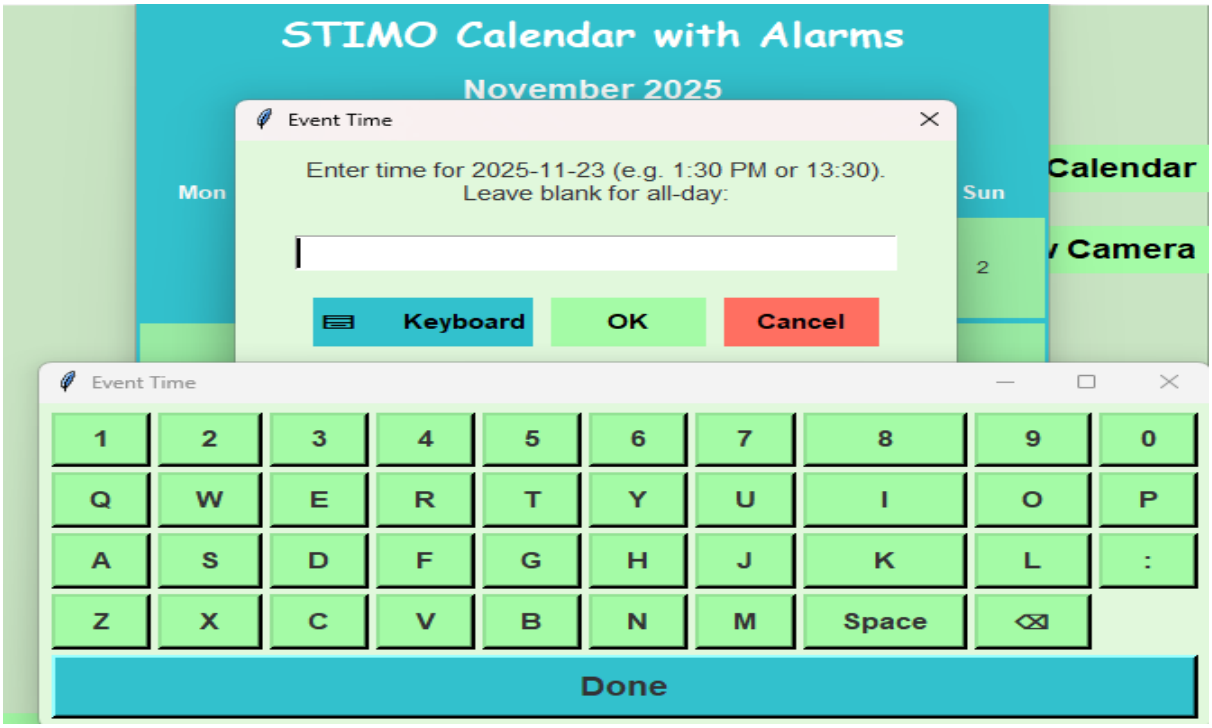


Figure 9.4 Select the add event with time

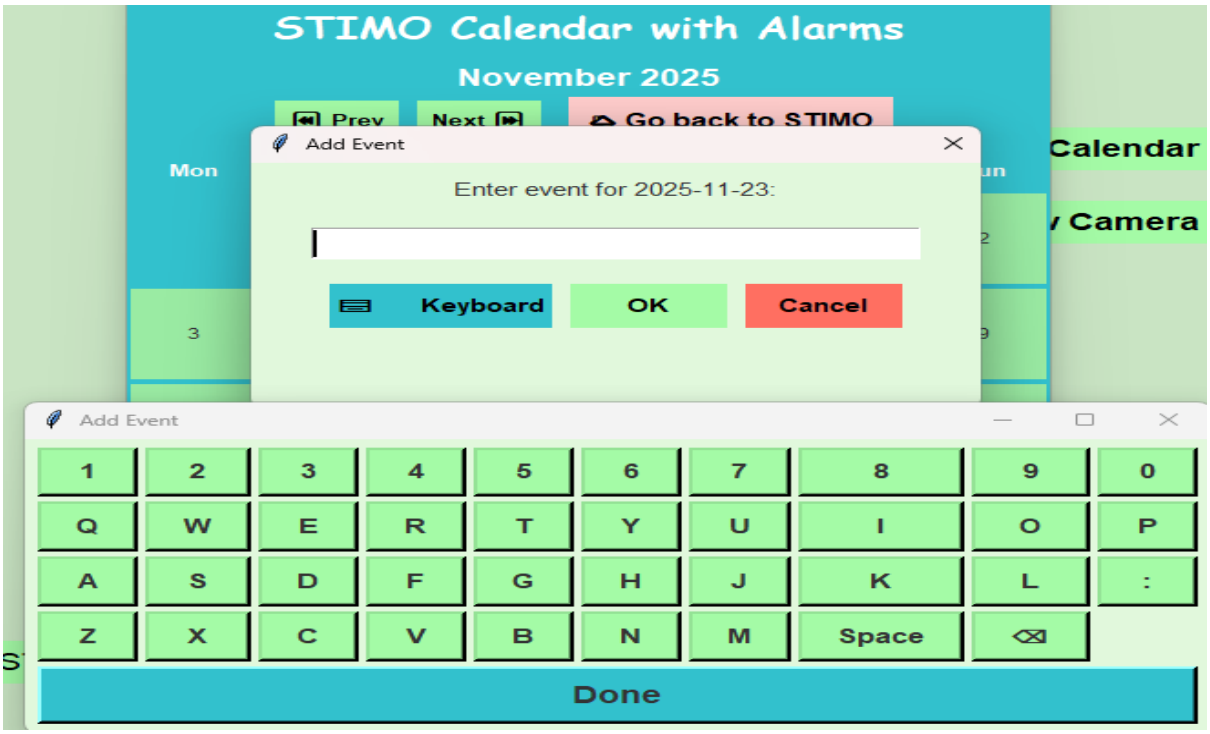


Figure 9.5: Enter the event

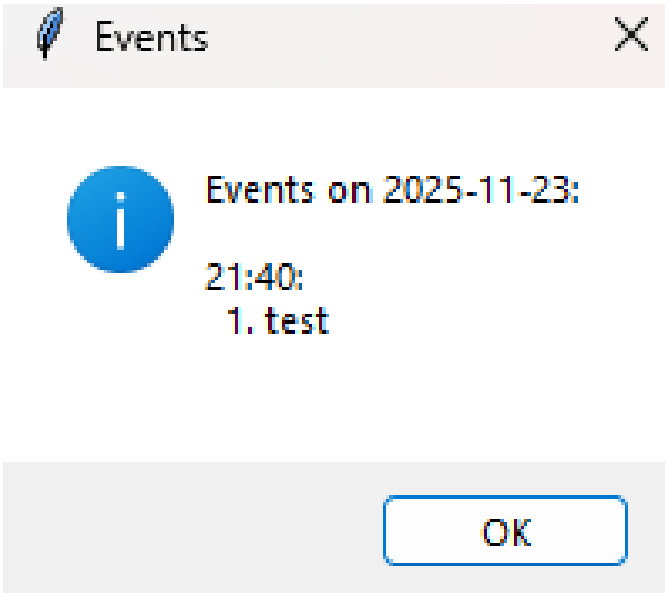


Figure 9.6: Done Input event

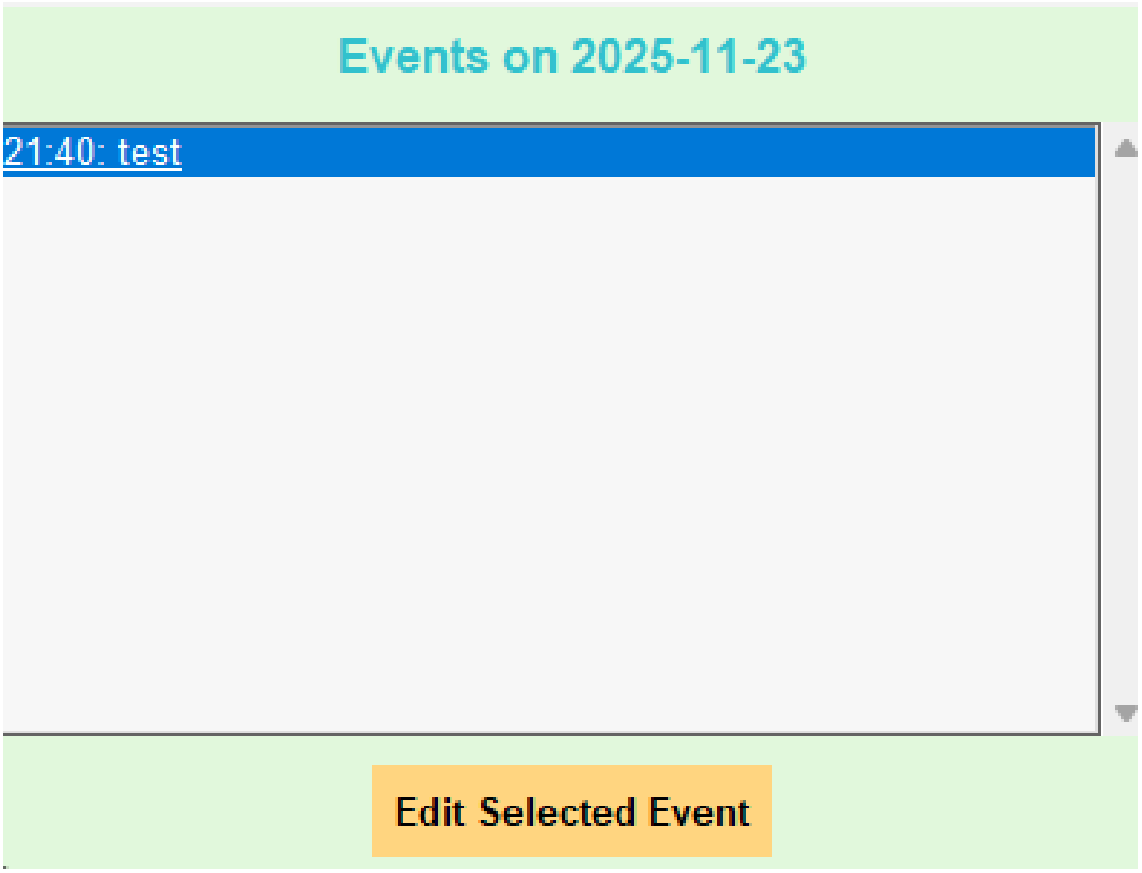


Figure 9.7: Edit Selected Event

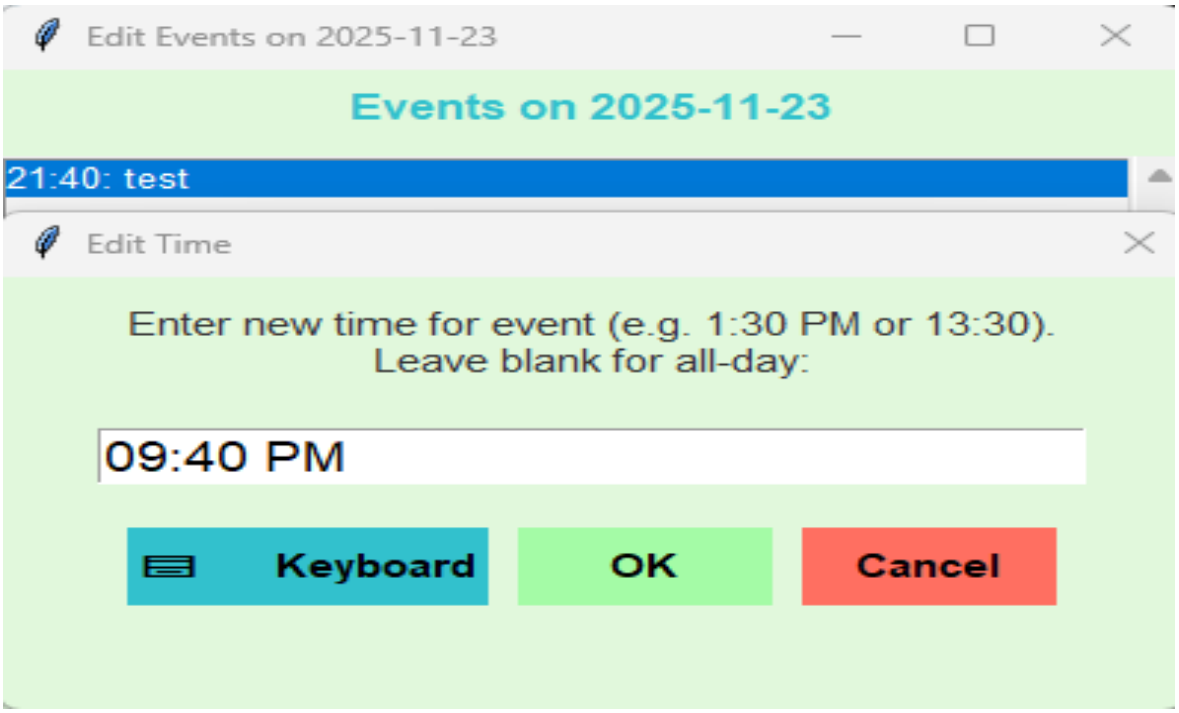


Figure 9.8: Enter new time

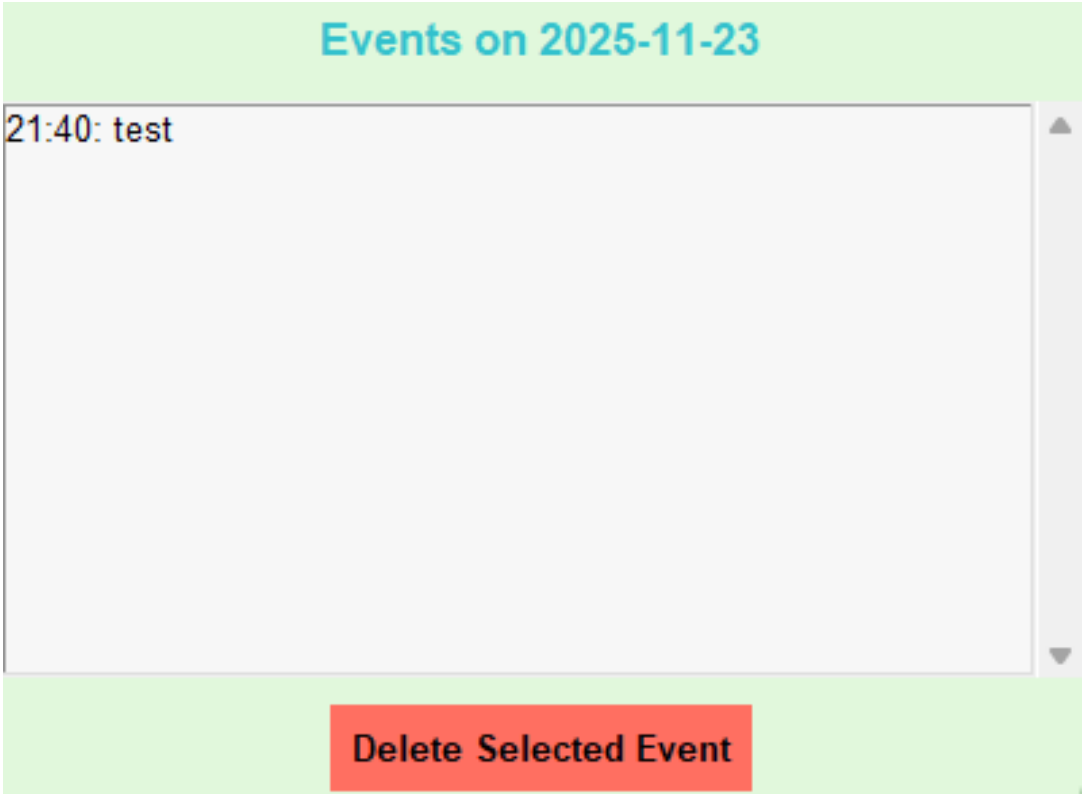


Figure 9.9: Delete Selected Event

Game Mode

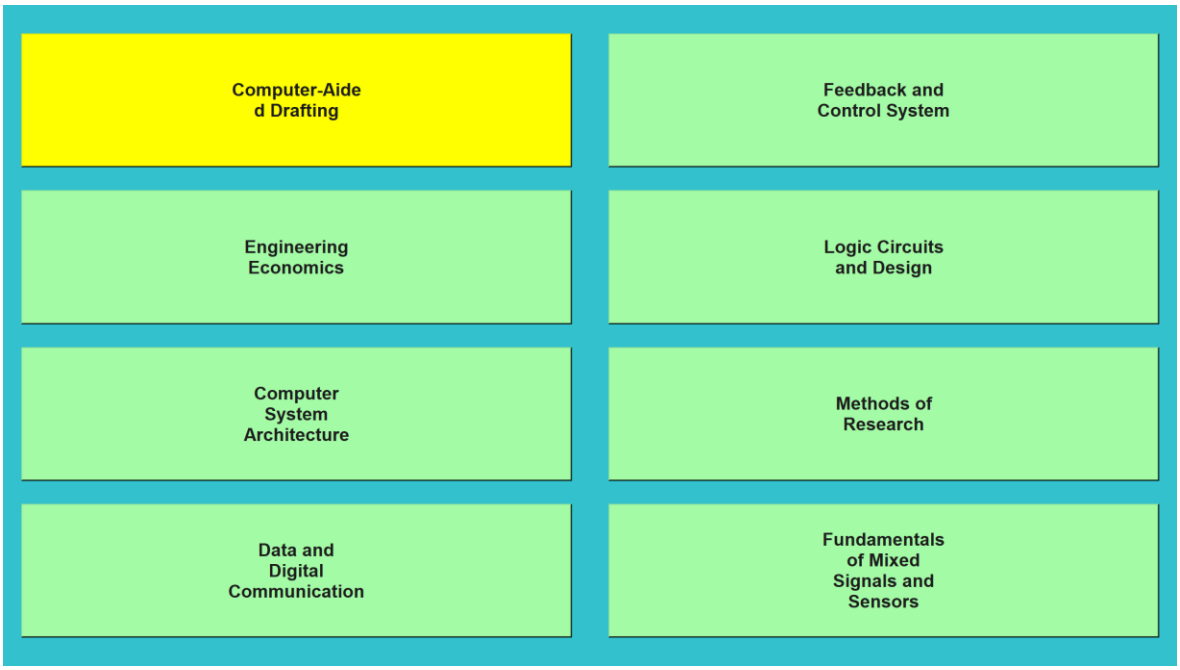


Figure 10: Select the Subject

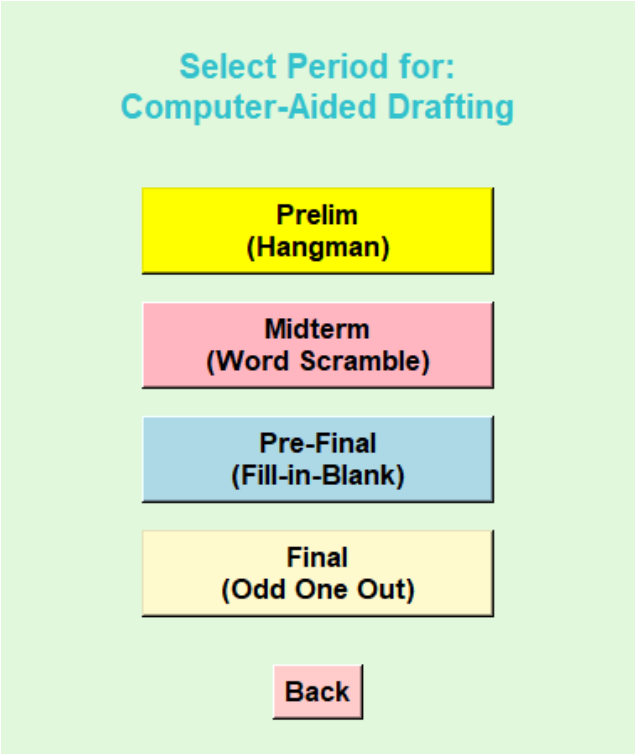


Figure 10.1: Select Perriod for Prelim, Midterm, Pre-final , Final

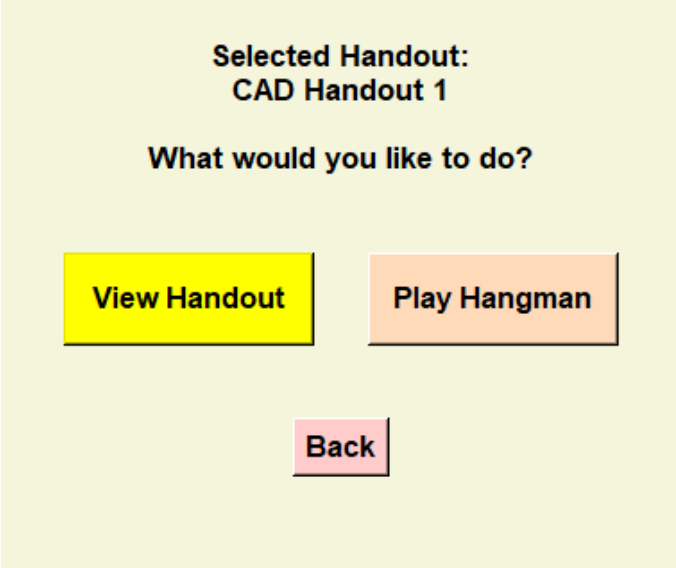


Figure 10.2: select View Handout or Play Games

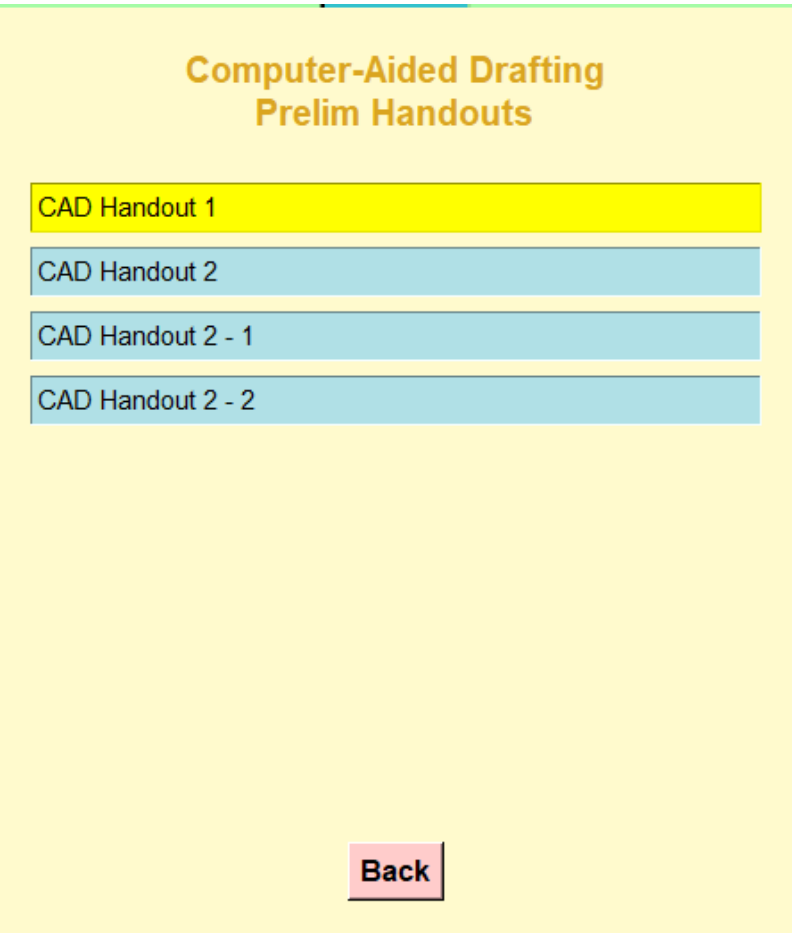


Figure 10.3: Select the Handouts Prelim,Midterm,Pre-final,Final

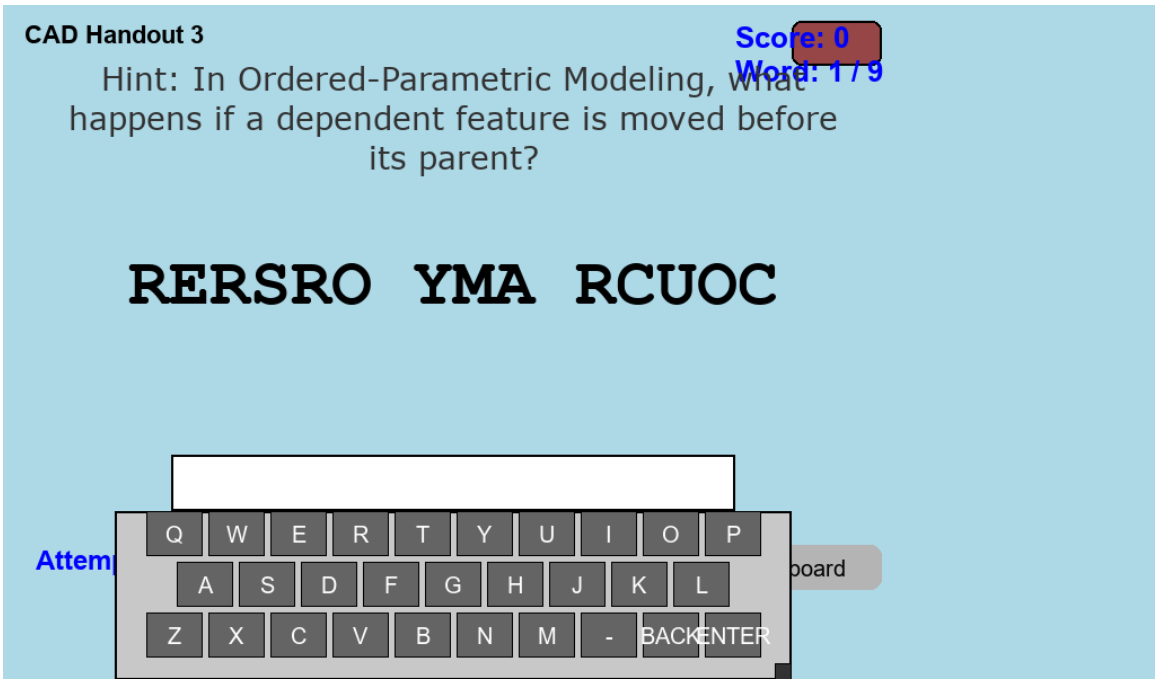


Figure 10.4: Play Games

PC Mode

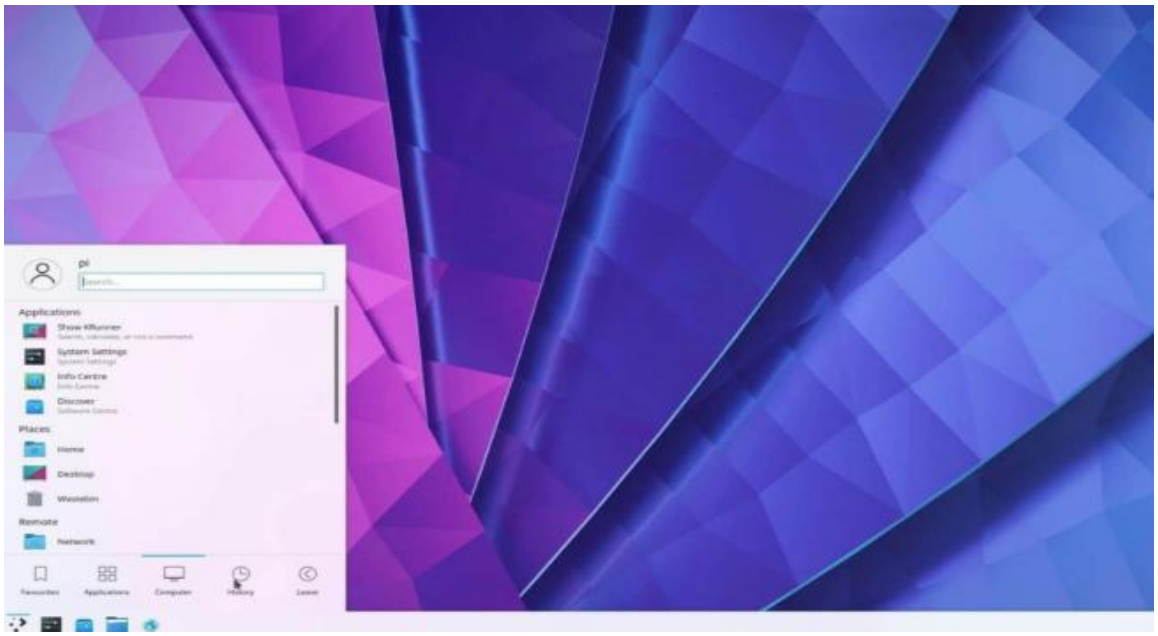


Figure 11: PC Mode

Hardware/Software

Hardware



Figure 12:

Raspberry Pi 4: *The Raspberry Pi 4 is a compact, energy-efficient single-board computer designed for versatile computing tasks such as programming, multimedia processing, and educational applications. In STIMO, it serves as the central processing unit, managing all mode-specific programs, animations, speech processing, and graphical user interface rendering. By handling the main software operations and multitasking functions, the Raspberry Pi 4 ensures smooth and responsive performance across STIMO's interactive features, while low-level hardware tasks—such as servo control and button input—are managed by the Raspberry Pi Pico and the PCA9685 module.*

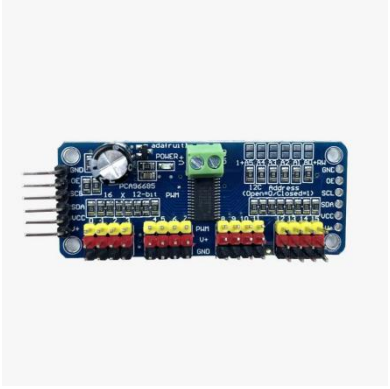


Figure 13:

PCA9685 PWM Servo Drive*r: The PCA9685 PWM Servo Driver provides precise control of the MG90S servo motors through I²C communication with the Raspberry Pi 4. Meanwhile, all button input handling is offloaded to the Raspberry Pi Pico, which manages mode switching and command signals. This separation of tasks ensures faster response times and a more stable, fully integrated control system.*



Figure 14:

APROTII 7-inch Touchscreen: *The APROTII 7-inch touchscreen is widely used in various applications for interactive displays. In STIMO, it serves as the main interface for visual outputs, including animated facial expressions and system interactions, while also enabling users to navigate and interact with the calendar and other functions through touch.*



Figure 15:

R305 Fingerprint Sensor: *The R305 fingerprint sensor is commonly used for biometric authentication in various security systems. In STIMO, it provides secure user access by verifying fingerprints, adding an additional layer of security.*



Figure 16:

MCU-Based Push Buttons: *MCU-based push buttons are commonly used for simple user input in electronic systems. In STIMO, these buttons enable intuitive mode switching, allowing users to easily toggle between STIMO Mode, Game Mode, and PC Mode without the need for complex inputs, ensuring a seamless and accessible user experience.*



Figure 17:

Creative Pebble 2.0 USB-Powered Speaker: *The Creative Pebble 2.0 USB-Powered Speakers are compact audio devices designed for high-quality sound in small spaces. In STIMO, these speakers provide clear and immersive audio output for responses, notifications, and media playback. Integrated into the system, they ensure that STIMO's*

verbal interactions and notifications are delivered with crisp, high-quality sound.



Figure 18:

The OPENMALL Portable Mini USB 2.0 Microphone: *The OPENMALL Portable Mini USB 2.0 Microphone is an omni-directional stereo microphone designed for high-quality audio capture. It connects directly to the Raspberry Pi via USB, providing clear and consistent voice input without requiring additional drivers or configuration. In STIMO, this microphone enables accurate speech recognition for hands-free operation, allowing users to issue voice commands seamlessly. Its compact and portable design makes it ideal for integration into the STIMO system, ensuring reliable performance and clear sound capture during interactive sessions.*



Figure 19:

DCDC-NUC Automotive Power Supply*: STIMO's portable operation. It accepts a wide input range of 6–48V and delivers a stable 5V DC output, which is required by the Raspberry Pi 4 single-board computer. This ensures efficient and consistent power delivery,*

allowing the Pi to maintain stable performance even during high-processing workloads. Its compact and lightweight design makes it easy to integrate into STIMO's hardware setup, improving power management and overall portability without compromising system performance.



Figure 20:

LiFePO4 6000mAh Battery Cell: *The 3 LiFePO4 battery provides a stable source with a 6000mAh capacity, offering safe and reliable energy for STIMO. Its high discharge rate and long cycle life ensure consistent performance during motor movements and system operations. Compact and durable, it helps maintain STIMO's portability and steady runtime.*



Figure 21:

MG90S Servo Motor: *The MG90S Servo Motor enables precise mechanical movements for STIMO's interactive functionalities. It actuates STIMO's limbs, allowing lifelike movements and physical responses to user commands. Integrated with the system's software, it ensures smooth*

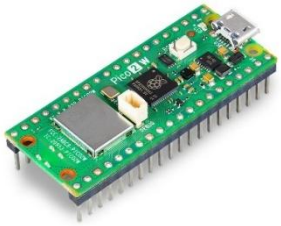


Figure 22:

The Raspberry Pi Pico: *The Raspberry Pi Pico serves as STIMO's dedicated microcontroller for handling all button inputs. It reads the signals from the push buttons in real time and sends the corresponding commands to the main Raspberry Pi. Offloading GPIO tasks to the Pico ensures faster input response, reduced processing load, and stable mode switching.*

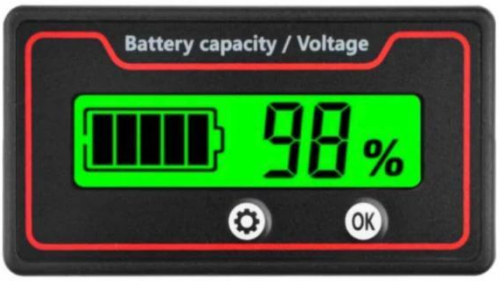


Figure 23:

Battery Capacity Monitor: *The battery capacity monitor displays the real-time charge level of STIMO's power source, helping users track remaining battery life. It provides quick visual feedback during operation and charging, ensuring the system is used safely and*

preventing unexpected shutdowns during activities.



Figure 24:

3D Printed Parts: *The 3D-printed parts form STIMO's custom body, housing, and mounting structures. These components provide a lightweight and durable frame that securely holds the Raspberry Pi, servos, buttons, and display. Using 3D printing allows precise fitting, easy customization, and rapid prototyping, ensuring the robot's mechanical design aligns perfectly with its internal hardware.*



Figure 25::

Centra Board: *The centra board serves as the main mounting platform for STIMO's internal components. It keeps the Raspberry Pi, Pico, power modules, and wiring*

organized and securely fixed in place. Its sturdy structure ensures stable hardware installation and helps maintain a clean, reliable layout inside the robot.



Figure 26:

USB Extension Cable: *The USB extension cable expands STIMO's connectivity by providing additional USB ports for external devices such as keyboards, mice, or storage media. This ensures flexible and reliable support for the system's input and output requirements.*

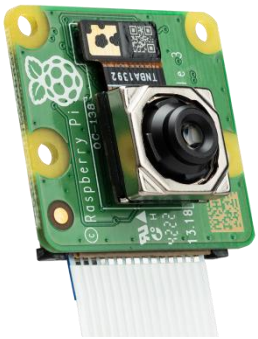


Figure 27:

Raspberry Pi Camera Module: *The Raspberry Pi Camera Module enhances STIMO by enabling real-time image capture and basic face-detection capabilities. When the camera detects the presence of a face, STIMO responds with a programmed movement or verbal greeting, improving its sense of interactivity and awareness. The module connects directly to the Raspberry Pi 4B through the CSI interface, ensuring fast and reliable data transfer.*

This camera integration provides STIMO with simple but effective visual responsiveness.



Figure 28:

Power Converter Super: *Utilizing a highly efficient buck converter maximizes STIMO's operating duration by minimizing power loss, a critical design choice that ensures the robot maintains peak performance throughout extended gamified learning sessions without requiring frequent recharges.*



Figure 29:

2Pin Snap-in ON/OFF KCD1-101: *The On and Off Switch is implemented as the primary physical control and safety mechanism for STIMO, allowing students to immediately power cycle or emergency shut down the system, which is essential for safely debugging code and managing the hardware during hands-on learning and gamification activities.*



Figure 30:

Portable Power Bank USB: *serves as the necessary high-capacity, regulated 5V power supply, fundamentally enabling STIMO's mobility and ensuring sustained power delivery to the Raspberry Pi 4B and peripherals, which is essential for the robot's functionality during untethered, interactive gamified learning task*s

Software:



Figure 31:

Raspberry Pi OS*: is the operating system that powers STIMO, providing the foundational environment for all applications and system functions. It manages multitasking between STIMO Mode, Game Mode, and PC Mode, ensuring smooth transitions and stable performance. Raspberry Pi OS supports the integration of external peripherals and offers a reliable platform for running STIMO's interactive features, including the calendar system, mini-games, and speech recognition components. It ensures that STIMO operates efficiently while maintaining an accessible and user-friendly interface for students*



Figure 32:

Python: (Pygame & Libraries): *Python is the primary programming language used to develop STIMO's software. It handles the core logic, including the processing of user inputs, managing system commands, and controlling STIMO's interactions. Python enables*

integration with hardware components like the servo motors, microphone, and touchscreen, while also supporting external libraries for functionalities such as speech-to-text, text-to-speech, and handling game logic. It allows for efficient, flexible development of STIMO's interactive features and ensures smooth communication between hardware and software.

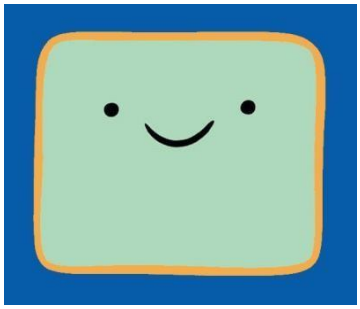


Figure 33:

STIMO Software: *STIMO Mode controls all the interactive features of the robot. It manages motorized movements using MG90S Servo Motors, audio through Creative Pebble 2.0 USB-Powered Speakers, and visual outputs. The software processes user commands through speech-to-text (STT) or keyboard input, triggering preprogrammed responses and movements.*

It provides text-to-speech (TTS) responses and animated facial expressions, with colors changing based on emotions (yellow for happy, blue for sad, red for angry). STIMO Mode also includes a calendar system for tracking homework and deadlines, sending reminders via TTS, animations, and movements.

The software ensures smooth operation, integrating all features for a seamless, interactive experience.

RESULTS AND DISCUSSIONS

The results of the study confirm the successful design and development of STIMO: An Interactive Robot for Learning and Gamification in BSCpE Education, meeting all established specific objectives through the integration of specialized hardware and customized Python-based software. The implementation follows a prototyping approach that involved iterative testing and refinement of the mechanical design, electronic circuitry, and programming logic to ensure fluid and reliable user interaction across its core functionalities.

The verification of the system's core features serves as the primary evidence, demonstrating how the specialized hardware components work in synergy with the three custom operating modes: STIMO Mode, Game Mode, and PC Mode. The specific objective to develop a robot that synchronizes motorized limb movements and preprogrammed responses with animations was achieved using MG90S Servo Motors, the PCA9685 PWM Driver Module, and the Raspberry Pi Pico. This resulted in real-time synchronized physical movements (e.g., kicking, waving) with Text-to-Speech (TTS) output and on-screen facial animations, which display three synchronized emotions (yellow for happy, blue for sad, red for angry). The objective to create a Game Mode that incorporates subject-specific mini-games was met through Python/Pygame software logic and a BSCpE 3-1 curriculum database, successfully implementing four term-specific games: Hangman (Prelim), Word Scramble (Midterms), Fill-in-the-Blank Challenge (Prefinals), and Odd One Out (Finals). The development of a calendar system utilized a Touchscreen LCD, Speech-to-Text (STT), and local database storage, allowing students to input and manage schedules, with reminders delivered via TTS, movements, and animations. Finally, the objective to implement push buttons for seamless switching was achieved using MCU-based Push Buttons connected to Raspberry Pi GPIO pins, which allows for instantaneous, hardware-driven switching between the three main modes.

System Design and Technical Verification

The physical construction and electronic system successfully validated the project's technical specifications. The design is centered on the Raspberry Pi 4 Model B, which functions as the central processing unit, managing the system's three core modes, graphical user interface (GUI), and speech processing. Sample raw data for the system's hardware specification is documented in the full technical documentation. The Raspberry Pi Pico is dedicated to handling low-level tasks, specifically the input signals from the push buttons, which enables fast and stable mode switching.

A crucial aspect of the design verification is the separation of high-level and low-level control. The PCA9685 PWM Servo Driver manages the MG90S Servo Motors via I^2C communication, allowing the Raspberry Pi 4 to focus on software execution, thereby ensuring stable and fluid coordinated movements for STIMO Mode.

The system’s three distinct operating modes provide a holistic approach to student interaction. STIMO Mode is the primary interactive assistant, providing text-to-speech responses and personalized schedule reminders. Game Mode is the gamified learning platform, and PC Mode transforms the unit into a functional Raspberry Pi OS desktop for productivity. Furthermore, the system includes security verification via the R305 fingerprint sensor, which runs an independent standalone program that must be successfully authenticated before accessing STIMO’s functions.

Discussion of Gamification and Learning Effectiveness

The implementation of the Game Mode provides tangible evidence for the project's focus on gamification as a tool for enhanced learning engagement, supporting the finding that 67% of students report improved motivation with interactive, gamified elements. The design moves beyond generic quizzes by linking four distinct mini-games to the specific grading periods (Prelim, Midterms, Prefinals, Finals) of the BSCpE 3-1 curriculum, ensuring that the learning is both engaging and highly relevant to the student's current academic modules.

For example, when a user selects a subject and the Prefinals term, they are presented with the Fill-in-the-Blank Challenge. This game structure enforces review of specific handouts and topics for that period in an interactive format, transforming traditional, passive studying into an active, enjoyable experience. This course-specific integration sets STIMO apart from more general educational platforms.

Constraints and Future Implications

While the system successfully achieved its developmental objectives, limitations were identified and verified during testing. The reliance on a Raspberry Pi 4 means the robot lacks the processing power for highly demanding software, such as advanced 3D modeling tools. Additionally, certain PC Mode functionalities are dependent on a stable internet connection for accessing online resources. The portability of STIMO is constrained by its current power source; the 6000mAh power bank provides a limited battery runtime of approximately 1 hour of continuous operation. These constraints provide a foundation for future development iterations, such as upgrading the CPU and battery system to improve computational performance and operational duration.

CONCLUSION

Based on the accomplishments of this study, the following conclusions are established:

The successful integration and synchronization of the Raspberry Pi 4 (software/TTS) with the Raspberry Pi Pico and servo motors (hardware/movement) validates the technical feasibility of creating a cohesive, multi-modal robotic assistant that can deliver engaging, coordinated physical and verbal interactions in an educational setting.

The implementation of the Game Mode, featuring curriculum-aligned mini-games, confirms that gamification is an effective strategy to transform traditional BSCpE 3-1 curriculum review into an enjoyable, relevant, and interactive learning experience, which addresses the gap of unengaging study methods.

The developed Calendar System serves as a practical, automated organizational tool that directly supports student productivity and schedule management by providing timely, personalized reminders through multi-modal notifications, mitigating the risk of missed deadlines.

The integration of MCU-based push buttons provides a highly intuitive and accessible interface for seamless switching between STIMO’s modes, ensuring a stable and user-friendly multitasking environment where the core assistant functions are consistently available to the user

Recommendations

Based on the conclusions and the noted limitations of the study, the following recommendations are proposed:

System Enhancement: To address the limitations of processing power and limited battery runtime, future research should explore upgrading the central Single Board Computer (SBC) to a higher-performance model and integrating a higher-capacity power source to extend the robot's operational duration for long study sessions.

Functionality Expansion: The curriculum-specific content in the Game Mode should be expanded to cover additional year levels and courses beyond the BSCpE 3-1 program to maximize the robot's educational utility. Additionally, developers should focus on creating more robust offline capabilities for the core STIMO Mode and Game Mode to reduce dependency on a constant internet connection.

Future Research and Evaluation: It is recommended that a subsequent study be conducted to perform a formal, large-scale acceptability and impact evaluation of STIMO involving a broader population of students. This will empirically measure the system's long-term effectiveness on student motivation, engagement, and academic perform.

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APPENDICES

APPENDIX A. GANTT CHART

MONTH / ACTIVITY	JUNE				JULY				AUGUST				SEPTEMBER				OCTOBER				NOVEMBER				DECEMBER				JANUARY				FEBRUARY				MARCH			
Project Proposal												x	x	x	x	x																								
Data Gathering												x	x	x	x	x																								
Analyzing the Study													x	x	x	x	x																							
Identifying the Hardware														x	x	x	x																							
Identifying the Software														x	x		x	x					x	x																
Purchasing the Materials																x	x	x	x																				
Designing the Circuit																		x	x	x																			
Prototype Building																				x	x	x	x																
Checking the System																					x	x	x																
Testing the Prototyp																					x	x																	
Finalizing the Documentation															x	x	x	x	x	x	x		x																

APPENDIX B. ACTUAL THESIS EXPENSES

THESIS EXPENSES

Quantity	Specifics	Approximate Cost	Actual Cost
1	Battery Capacity Indicator 12V	PHP 276	PHP 276
1	Touchscreen(APROTII 7-inch 1024x600 IPS Portable Monitor)	PHP 2,600	PHP 2,600
1	Rasberry Pi 4b	PHP 5,999	PHP 5,999
1	Raspberry pi Camera Module	PHP 1,200	PHP 1,200
1	3D Printed Parts	PHP 1,000	PHP 1,000
1	Fingerprint Sensor	PHP 800	PHP 800
10	Push Buttons	PHP 10	PHP 100
1	OPENMALL Portable Mini USB 2.0 Microphone	PHP 200	PHP 200
4	MG90S Servo Motor	PHP 170	PHP 680
1	Creative Pebble 2.0 USB-Powered Speaker	PHP 300	PHP 300
4	USB Extension	PHP 150	PHP 600
1	Raspberry Pi Pico	PHP 690	PHP 690
4	DC-DC Buck Converter	PHP 187.5	PHP 750
1	Power Bank	PHP 200	PHP 200
1	DC-DC Automotive Power Supply	PHP 103	PHP 103
4	LiFePo4 6,000 mah Battery	PHP 90	PHP 360
1	Centra Board	PHP 800	PHP 800
1	2Pin Snap-in ON/OFF KCD1-101	PHP 30	PHP 30
1	PWM Servo Driver	PHP 220	PHP 220
			Total: 16,908

Prepared by:

Elias John R. Dorado **Jeremy Brain D. Rentillo**

Gabriel B. Tosco

Noted by:

Erlinda R. Dorado **Norly D. Rentillo**

Jennifer B. Tosco

Approved by:

ENGR. KENT IAN S. DAVID MR. KENDRICK YUPANO

APPENDIX C. USER'S MANUAL

Appendix C. User's Manual

This manual provides comprehensive instructions for the operation and management of STIMO: An Interactive Robot for Learning and Gamification in BSCpE Education. STIMO is a standalone educational robot designed to assist students through interactive learning, gamified activities, schedule management, and simple computer operations. Powered by a Raspberry Pi 4, STIMO integrates speech technology, servo-based movements, touchscreen controls, camera-based interaction, and MCU-driven input buttons.

By utilizing STIMO's built-in touchscreen interface, voice recognition, physical buttons, mini-games, and calendar system, users can navigate various modes, receive reminders, play curriculum-based games, and access a Raspberry Pi desktop environment.

Primary System Capabilities

STIMO Mode (Interactive Assistant)

- *Displays animated facial expressions synchronized with movements.*
- *Recognizes voice commands through STT (speech-to-text).*
- *Uses servo motors for gestures such a waving and kicking.*
- *Manages schedules with a built-in calendar sends reminders.*

Game Mode (Gamified Learning)

- *Offers subject-specific mini-games aligned with the BSCpE 3-1 curriculum.*
- *Includes Hangman, Word Scramble, Fill-in-the-Blank, and Odd One Out.*

- *Allows users to select subjects and grading periods.*
- *Enhances lesson retention through interactive gameplay.*

PC Mode (Raspberry Pi Desktop)

- *Converts STIMO into a functional computer using Raspberry Pi OS*
- *Allows file management, browsing, and application access.*
- *Keeps STIMO Mode running in the background for voice or button input.*

Security & User Access

- *Features fingerprint authentication using the R305 sensor.*
- *Utilizes a standalone verification system independent of the OS login.*

Visual Interaction

- *Integrates a Raspberry Pi Camera Module for detecting faces and presence.*
- *Provides greeting responses and simple animations based on detection.*

System Architecture

- **Hardware Control Unit:** *Includes Raspberry Pi 4, Raspberry Pi Pico, PCA9685, MG90S servos, 7-inch touchscreen, microphone, speakers, fingerprint sensor, LiFePO4 battery, DC converters, MCU buttons, and camera module*
- **Software System***: Python, Pygame, STT, TTS, SQLite, and Raspberry Pi OS*

coordinate STIMO's modes and interactions.

Starting STIMO (How to Turn On and Begin Using the Robot)

1. Switch On the Power

Toggle the KCD1-101 ON/OFF power switch located on STIMO's chassis. This supplies power to the Raspberry Pi 4, Raspberry Pi Pico, servos, and connected modules.

2. System Boot

*After powering on, the **Raspberry Pi OS will automatically boot**. You will see the screen light up and load the initial startup processes.*

3. Fingerprint Authentication

Before accessing any STIMO features, the R305 fingerprint sensor activates automatically.

- *Place your registered finger on the scanner.*
- *If the fingerprint matches, the system unlocks and continues.*
- *If authentication fails, STIMO will prompt you to try again.*

4. STIMO Launcher Startup

Once authentication is successful, the STIMO Launcher appears on the touchscreen.

This launcher is the central hub where users select the robot's operating mode.

5. Choose a Mode Using Buttons or Touch

STIMO provides three main operational modes:

- *Mode 1 — STIMO Mode (Interactive Assistant)*
- *Mode 2 — Game Mode (Gamified Learning)*
- *Mode 3 — PC Mode (Raspberry Pi Desktop)*

Users can select a mode using either:

The physical MCU mode buttons, or

Touchscreen-based selection (if implemented on your interface)

6. Mode Activation

After selecting a mode, STIMO will automatically load the corresponding program:

- *STIMO Mode → Animated assistant with voice recognition, gestures, and reminders*
- *Game Mode → Educational mini-games*
- *PC Mode → Full Raspberry Pi desktop while STIMO continues listening in the background*

7. STIMO is Ready for Use

Once the mode is loaded, STIMO becomes fully operational. Users may interact through:

- *Voice commands (STT)*

- *Touchscreen controls*
- *Physical buttons*
- *Calendar input and schedule tracking*
- *Facial detection reactions*

Troubleshooting

- **No Audio:** *Check the speaker USB connection.*
- **No Response to Buttons:** *Restart the Raspberry Pi Pico.*
- **Servos Not Moving:** *Check PCA9685 wiring.*
- **Fingerprint Not Detected:** *Clean the sensor surface.*
- **Camera Not Detecting Face:** *Ensure proper lighting.*

Safety Instructions

- *Do not expose to moisture.*
- *Use only recommended 5V power source.*
- *Avoid pulling servo limbs manually.*
- *Do not block ventilation areas of the Raspberry Pi.*

APPENDIX D. CURRICULUM VITAE OF RESEARCHERS

Curriculum Vitae of

Elias John R. Dorado

Camanci Sur, Numancia, Aklan
Elias.dorado90@gmail.com
09151356028

EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary	N/A	STI College Kalibo
TechVoc	N/A	
High School	March 2018	Panay Technological College
Elementary	March 2014	New Buswang Elementary School

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience/ Job Title	Name and Address of Company or Organization
month year		
month year		
month year		
month year		

Listed in reverse chronological order (most recent first).

AFFILIATIONS

Inclusive Dates	Name of Organization	Position
month year		
month year		
month year		
month year		

Listed in reverse chronological order (most recent first).

SKILLS

SKILLS	Level of Competency	Date Acquired
		month year
		month year
		month year

TRAININGS, SEMINARS OR WORKSHOP ATTENDED

Inclusive Dates	Title of Training, Seminar or Workshop
month year	
month year	
month year	
month year	

Listed in reverse chronological order (most recent first).

Curriculum Vitae of

Jeremy Brian D. Rentillo

Mabilo, Kalibo, Aklan
Jeremybrianrentillo@gmail.com
09948471047

EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary	N/A	STI College Kalibo
TechVoc	N/A	
High School	March 2020	Saint Gabriel College
Elementary	March 2014	Aklan Catholic College

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience/ Job Title	Name and Address of Company or Organization
month year		
month year		
month year		
month year		

Listed in reverse chronological order (most recent first).

AFFILIATIONS

Inclusive Dates	Name of Organization	Position
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SKILLS

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TRAININGS, SEMINARS OR WORKSHOP ATTENDED

Inclusive Dates	Title of Training, Seminar or Workshop
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month year	
month year	
month year	

Listed in reverse chronological order (most recent first).

Curriculum Vitae of

Gabriel B. Tosco

Afga, Tangalan, Aklan
Gabriel.Tosco240@gamil.com
09215127244

EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary	N/A	STI College Kalibo
TechVoc	N/A	
High School	March 2020	Tangalan, National High School
Elementary	March 2014	Poblacion , Tangalan , aklan

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience/ Job Title	Name and Address of Company or Organization
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month year		
month year		
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AFFILIATIONS

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Listed in reverse chronological order (most recent first).

SKILLS

SKILLS	Level of Competency	Date Acquired
		month year
		month year
		month year

TRAININGS, SEMINARS OR WORKSHOP ATTENDED

Inclusive Dates	Title of Training, Seminar or Workshop
month year	
month year	
month year	
month year	

Listed in reverse chronological order (most recent first).

ADVISER'S ACCEPTANCE FORM

NAME OF PROPONENTS: **Elias John R. Dorado**
Jeremy Brian D. Rentillo
Gabriel B. Tosco

APPROVED RESEARCH TITLE: STIMO: An Interactive Robot for Learning and Gamification in BSCpE Education

AREA OF STUDY: **Bachelor of Science in Computer Engineering**

CONFORME:

Engr. Kent Ian S. David Thesis

APPROVED BY:

Ms. Maharlika A. Andrade **Date: 11 October, 2024**
Thesis Coordinator

NOTED BY:

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